If You Can't Break It, Keep It: Choiceless Computation under Syntactic Symmetry

Abstract

Sinitic adjectival coordinatives systematically exclude the simplex *AB form while requiring reduplicated outputs AABB or ABAB. I argue that this gap follows from a restriction on deterministic evaluation over symmetric domains, formalized as the Choiceless Computation Theorem (CCT). The CCT recasts Chomsky's (2013, 2015) labeling theory in automorphism-theoretic terms, adds a fourth case (the Null Label), and yields a falsifiable 2×2 typology unifying AABB distributive exponence, ABAB pied-piping, ATB extraction, and Right Node Raising. On this view, narrow syntax is crash-tolerant: symmetry need not be broken within NS, but can persist and be rendered legible at the interfaces.

Keywords: Automorphism, Choiceless Computation Theorem, Crash-tolerant Syntax, Distributive exponence, Labeling Algorithm, Syntactic symmetry

'lorsque certaines causes produisent certains effets, les éléments de symétrie des causes doivent se retrouver dans les effets produits'

— Pierre Curie

1 The Symmetry Problem at the Merge-Interface Boundary

Under the Strong Minimalist Thesis (SMT), the computational system of human language (C_{HL}) operates with the simplest structure-building operation, Free-Merge (Chomsky 2004 et seq.). When Merge combines two minimal elements, the result is an unordered set $\{\alpha, \beta\}$ that does not inherently designate a projecting head. As noted by Moro (2000) and Moro & Roberts (2023), this inevitable first step of any derivation, the *ignition* problem, gives rise to a symmetric configuration.

In its most reduced form, the combination of two acategorial roots yields the most economical Syntactic Object (SO). Yet such root symmetry creates a fundamental tension with the legibility conditions imposed at the interfaces. I argue that this tension drives a systematic and previously underappreciated class of post-syntactic phenomena.

Consider the following paradoxical pattern in Sinitic languages, where the concatenation of monosyllabic gradable adjectives systematically excludes the morphophonologically simpler simplex *AB form, while strictly requiring reduplicated outputs such as AABB or ABAB. The Mandarin contrast in (1) illustrates the pattern.[1] Notably, even the presence of *hen* fails to rescue the simplex form.[2]

(1) a. * *zhe-ge baomihua hen* [*tian-xian* / *xian-tian*] *de*
this-CL popcorn very sweet-salty salty-sweet DE

b. *zhe-ge baomihua* [*tian-tian-xian-xian* / *xian-xian-tian-tian*] *de*
this-CL popcorn sweet-sweet-salty-salty salty-salty-sweet-sweet DE

c. *zhe-ge baomihua* [*tian-xian-tian-xian* / *xian-tian-xian-tian*] *de*
this-CL popcorn sweet-salty-sweet-salty salty-sweet-salty-sweet DE
'The popcorn is sweet and salty.'

The pattern in (1) runs systematically counter to expected grammatical economy. The AABB forms in (1b) heavily violate the Obligatory Contour Principle (OCP) by placing identical syllables with identical tones in adjacent positions (McCarthy 1986; Yip 1988), yet they remain the only convergent options. At the same time, the freely reversible ordering of the two adjectives, together with a stable conjunctive interpretation $P \wedge Q$ and high productivity, argues against an analysis in terms of fixed lexical compounds. Moreover, the absence of categorial and selectional restrictions on the input adjectives suggests that the relevant atoms enter the derivation as bare roots lacking categorial and formal features, a view consistent with Distributed Morphology (DM) (Marantz 2007; Acquaviva

[1] Cantonese judgments are based on the author's native intuitions; Mandarin judgments are based on the author's native intuitions and were additionally verified with native speakers at SICOGG 27.

[2] On the requirement that bare gradable adjectives in Sinitic predicative positions be licensed by degree morphology, see Liu (2010) and Grano (2012).

2009; De Belder & van Craenenbroeck 2015). Setting aside apparent counterexamples and alternative analyses, which §2 addresses systematically, the data point toward a symmetric Merge of two roots, forming the set $\{\sqrt{A}, \sqrt{B}\}$.

On the analysis developed below, the *AB gap can be derived from root symmetry without feature-driven machinery. The symmetric set $\{\sqrt{A}, \sqrt{B}\}$ lacks both hierarchical asymmetry and featural specification, and therefore cannot support the distinctions required for interpretation. This indeterminacy surfaces differently at the two interfaces, but in each case as a reflex of the same representational deficit. At the Sensorimotor (S-M) interface, mutual c-command generates contradictory precedence statements under the Linear Correspondence Axiom (LCA) (Kayne 1994; Malanoski & Kandybowicz 2026). At the Conceptual-Intentional (C-I) interface, the absence of categorical structure prevents compositional interpretation.

Standard approaches assume that such symmetry must be resolved within Narrow Syntax. One familiar strategy is Internal Merge, which breaks mutual c-command by displacing one element and restoring labelability through the remaining asymmetric configuration (Chomsky 2013, 2015; Citko 2018). Another appeals to feature sharing, whereby the shared feature of two XPs determines the label of their symmetric combination (Chomsky 2013). In bare-root coordination, however, both strategies are unavailable as they presuppose functional structure and feature-driven computation. The relevant roots provide neither shared features nor probe-goal asymmetries for the syntax to exploit.

I propose instead that the symmetric set $\{\sqrt{A}, \sqrt{B}\}$ is transferred intact to the external components, where PF and LF apply their constraints independently. I will argue that PF converges on reduplication as the only convergent externalization of a symmetric structure, and that LF converges on conjunction as the only compositional parse compatible with the absence of hierarchical asymmetry. The result is a crash-tolerant view of syntax. Narrow Syntax generates structural possibilities without evaluating their inter-

pretability; symmetry may therefore persist beyond syntax and be rendered legible only through post-syntactic interface conditions.

What remains to be stated is the restriction that determines how such post-syntactic evaluation proceeds. The formal apparatus developed in this paper is the Choiceless Computation Theorem (CCT, §3), which characterizes how deterministic algorithms behave over symmetric inputs. In essence, the theorem establishes that no such algorithm can selectively target one element over another within a symmetric domain; any well-defined output must treat the symmetric elements uniformly.

Applied to Chomsky's (2013, 2015) labeling theory, the CCT provides an algebraic underpinning for the three established cases and adds a previously undiscussed fourth case, the Null Label. Applied more broadly, it yields a falsifiable 2×2 typology of how grammar handles structural symmetry, bringing AABB and ABAB reduplication into the same space as ATB extraction and Right Node Raising.

§2 isolates the root-symmetric empirical domain in Sinitic through five converging diagnostics. §3 formalizes the CCT and derives the Null Label as a fourth labeling case. §4 derives the attested resolutions of symmetric input and situates them within the 2×2 typology, extending to ATB and RNR as canonical calibration cases. §5 turns to the interpretive side, asking what the CCT permits at LF and why conjunction, rather than an asymmetric compositional parse, survives.

2 Not &P, Not aPs, Only Roots

2.1 Defining the Coordinatives

The target of inquiry is *true coordinatives*: dynamically generated structures defined by (i) the absence of hierarchical asymmetry and (ii) strict compositionality.

True coordinatives exhibit free linear ordering without truth-conditional change (Citko 2011), indicating the absence of hierarchical asymmetry:

(2) a. *tian-tian-xian-xian* ↔ *xian-xian-tian-tian*
sweet-sweet-salty-salty ↔ salty-salty-sweet-sweet
'sweet and salty' (Simultaneous State)

b. *tian-xian-tian-xian* ↔ *xian-tian-xian-tian*
sweet-salty-sweet-salty ↔ salty-sweet-salty-sweet
'alternatingly sweet and salty' (Alternating Aspect)

This ordering freedom is not reducible to information structure as the construction resists focal decomposition and cannot felicitously answer questions targeting a single property.[3] This contrasts with canonical endocentric compounds (e.g., German *blau-grün* 'bluish green' vs. *grün-blau* 'greenish blue'), where ordering is fixed and semantically consequential.

Second, true coordinatives encode strictly compositional meaning. Describing a zebra as *hei-hei-bai-bai* ('black-black-white-white') entails that both properties are independently instantiated across spatial parts rather than blended into a new property like 'grey':

(3) *banma gen ni yiyang, shen-shang dou hei-hei-bai-bai de*
zebra with you same body-on all black-black-white-white DE
'Zebras, like you, all have black and white (color) on their bodies.'

This compositional transparency distinguishes true coordinatives from lexicalized compounds such as English *bittersweet* or Finnish *sinivihreä* (blue-green, 'cyan'), which encode non-compositional or fused meanings (Renner 2008; Ralli 2019).

2.2 Against aP Analyses

Three alternatives to the root-symmetric analysis merit consideration: (i) a standard &P analysis (Munn 1993), conjoining two categorized aPs under a null conjunction head; (ii) a Pair-Merge analysis (Chomsky 2004; Bošković 2020b; Pan & Du 2024), forming an ordered pair $\langle a\text{P}, a\text{P}\rangle$; and (iii) an n-ary operation designed for unbounded coordination,

[3] Using standard question–answer congruence diagnostics (Rooth 1992), both orders degrade in contexts targeting only one conjunct (*zhe-ge baomihua tian-bu-tian*? 'Is this popcorn sweet?' — #*xian-xian-tian-tian de*). Similarly, disjunctive questions ($P \vee Q$) are incompatible with the conjunctive form. The construction encodes a single, inseparable conjunctive object. Ordering variation reflects an underlying symmetric representation, not discourse-related operations.

such as Chomsky's (2024) FormSet ($\{X_1, \ldots, X_m\}$). The following diagnostics systematically scrutinize these alternatives.

Strict Binarity. Sinitic coordinatives permit exactly two roots. A third element is strictly prohibited (4b), unlike overt *you...you...* coordination (4a):

(4) a. *zhe-dao cai you suan you tian you la.*
this-CL dish and sour and sweet and spicy
'This dish is sour, sweet, and spicy.'

b. *zhe-dao cai* * *suan-suan-tian-tian-la-la* / *
this-CL dish sour-sour-sweet-sweet-spicy-spicy
suan-tian-la-suan-tian-la de.
sour-sweet-spicy-sour-sweet-spicy DE
Intended: 'This dish is sour, sweet, and spicy.'

This restriction follows from binary Set-Merge (Chomsky 1995, 2000). Introducing a third element would force recursive Merge $\{\sqrt{C}, \{\sqrt{A}, \sqrt{B}\}\}$, creating an asymmetric hierarchy in which the symmetry-related interface effects derived here would not arise. This binarity straightforwardly rules out n-ary formation operations such as FormSet (Chomsky 2024), which by construction permits unbounded arity. Pair-Merge is likewise incompatible with the present analysis. Beyond violating the SMT adopted here, the conjunctive reading proposals for Pair-Merge crucially rely on shared categorial labels that acategorial roots cannot supply (see §3.2, §4 for further structural arguments).

Opacity to Internal Modification. aPs allow degree modifiers. Overt coordination permits internally modified conjuncts (5a). Yet, coordinatives strictly prohibit internal functional material (5b–c):

(5) a. *zhe-ge baomihua you* [$_{\text{DegP}}$ *hen tian*] *you* [$_{\text{DegP}}$ *hen xian*]
this-CL popcorn and very sweet and very salty
'This popcorn is both very sweet and very salty.'

b. * *zhe-ge baomihua hen-tian-hen-tian-hen-xian-hen-xian de*
this-CL popcorn very-sweet-very-sweet-very-salty-very-salty DE

c. * *zhe-ge baomihua hen-tian-hen-xian-hen-tian-hen-xian de*
this-CL popcorn very-sweet-very-salty-very-sweet-very-salty DE
Intended: 'This popcorn is both very sweet and very salty.'

Under the bare root analysis, the atomic units are acategorial roots $\{\sqrt{A}, \sqrt{B}\}$ merged prior to categorization. No functional structure is available for modification.

Prosodic Size Restrictions. The construction is strictly limited to monosyllabic bases. Disyllabic adjectives are acceptable as conjoined aPs (6a) but cannot enter the coordinative pattern (6b):

(6) a. *zhe-duo hua you xianyan you meili*
this-CL flower and bright and beautiful
'This flower is both bright and beautiful.'

b. *zhe-duo hua* * *xianxianyanyan-meimeilili* / *
this-CL flower bright-bright-beautiful-beautiful
xianyan-xianyan-meili-meili / * *xianyan-meili-xianyan-meili de*
bright-bright-beautiful-beautiful bright-beautiful-bright-beautiful DE
Intended: 'This flower is both bright and beautiful.'

The restriction tracks the root/aP distinction. Fully categorized aPs are prosodically unconstrained—mono-, di-, or polysyllabic—whereas bare roots in Sinitic are characteristically monosyllabic (Duanmu 2007; Xu 2008). Disyllabic forms like *xian-yan* are rather analyzed as morphologically complex aPs, not atomic roots (Zhang 2007; Hu & Perry 2018). The fact that the construction admits only monosyllabic elements aligns with a root-level derivation. Since NS has no access to phonological information (Chomsky 1995; Embick & Noyer 2001; Embick & Marantz 2008), the size restriction itself is evaluated at PF, but what it diagnoses is the categorial status of the merging elements.

Semantic Selection. The coordinative pattern applies almost exclusively to roots denoting directly perceptible, low-level sensory properties. Abstract or cross-dimensional combinations are excluded (7b), though they are freely coordinated phrasally (7a):

(7) a. *zhe-ge pingguo you hong you da*
this-CL apple and red and big

'This apple is both red and big.'

b. * *zhe-ge pingguo hong-hong-da-da de*
this-CL apple red-red-big-big DE
Intended: 'This apple is both red and big.'

This sensitivity to encyclopedic content is opaque to syntactic operations (Arad 2003; Embick & Marantz 2008) and reflects a condition on conceptual integrability at the C-I interface (developed in §5.3).

M-Word Integrity and *de* Distribution. The coordinative pattern forms a single Morphosyntactic Word (Lee-Kim 2017). Splitting into separate M-Words is prohibited, and a single *de* is required at the right edge (8a). Phrasal juxtaposition of independently reduplicated aPs, by contrast, requires *de* on each conjunct (8b):

(8) a. *zhe-ge baomihua* [*tian-tian-xian-xian* / *tian-xian-tian-xian*] *de*
this-CL popcorn sweet-sweet-salty-salty / sweet-salty-sweet-salty DE

b. *zhe-ge baomihua tian-tian de*, *xian-xian de*
this-CL popcorn sweet-sweet DE, salty-salty DE
'This popcorn is sweet, and salty.'

c. *zhe-ge baomihua tian-tian-(*de)-xian-xian-de*
this-CL popcorn sweet-sweet-DE-salty-salty-DE
Intended: 'This popcorn is sweet and salty.'

The obligatoriness of a single shared *de* in (8a) shows that the two roots share a single categorizer, providing further evidence for root-level derivation.

These diagnostics jointly establish that the atoms of the construction are acategorial sub-aP. One residual possibility remains at the diagnostic level: could a single root project a Root Phrase (√P) independently of a categorizer? This possibility is structurally foreclosed by the labeling architecture developed in (§3) and proven inapplicable.

2.3 Ruling Out Surface Look-alikes

Not all linearly concatenated adjectives in Sinitic instantiate symmetric root coordination. Two superficially similar cases systematically bypass the symmetry problem.

2.3.1 The Lexical Illusion

Non-compositional lexical compounds appear to challenge the ban on bare concatenation. Forms like *tian-suan-jiang* ('sweet-sour-sauce') surface as simplex AB:

(9) *zhe-ge jiang shi you tian you suan mei-cuo, dan ta bushi tian-suan-jiang*
this-CL sauce COP and sweet and sour not-wrong, but it NEG sweet-sour-sauce
'This sauce indeed tastes sweet and sour, but it is not the sweet-and-sour sauce.'

However, such forms exhibit fixed ordering (*suan-tian-jiang* denotes another distinct sauce), idiomatic use, and cannot be extended to arbitrary $P \wedge Q$ contexts. They pattern with stored lexical items (Borer 2013; Hu & Perry 2018), instead of dynamically generated coordinative structures. The same holds for frequent pairs like *suan-tian* ('sour-sweet') and *ma-la* ('numb-spicy'), which also exhibit frozen ordering (**tian-suan*, **la-ma*). Such simplex AB forms are considered *Listemes* (Di Sciullo & Williams 1987).

2.3.2 The Attributive Illusion

Bare concatenated adjectives AB appear acceptable in attributive contexts (10):

(10) *mei yao yi kou ji-rou, jiu liu-chu* [*tian xian de jiangzhi*].
every bite one CL chicken-meat, then flow-out sweet salty ATTR sauce
'With every bite of the chicken, sweet and salty sauce flows out.'

Two diagnostics show these are not root-level symmetric structures but paratactic adjunction of independent aPs. First, attributive modification allows open-ended iteration (11), unlike the strict binarity of root coordinatives:

(11) *ta yong tiaoliao-bao zuo-le yi-ge* [*tian la xian de jiangzhi*].
3SG use season-bag make-PFV one-CL sweet spicy salty ATTR sauce
'They made a sweet, spicy, and salty sauce using a seasoning packet.'

Second, each adjective can independently undergo canonical monosyllabic reduplication (12), generating an AA-BB-CC string that spans independent M-Words:

(12) *ta yong tiaoliao-bao zuo-le yi-ge* [*tian-tian la-la xian-xian*
3SG use season-bag make-PFV one-CL sweet-sweet spicy-spicy salty-salty
de jiangzhi].
ATTR sauce

'They made a (somewhat) sweet, spicy, and salty sauce using a seasoning packet.'

Thus, attributive modification involves paratactic adjunction of independently categorized aPs, each embedded within its own functional projection prior to concatenation. As the derivation never constructs the uncategorized symmetric object, attributive AB converges, not because symmetry is tolerated, but because symmetry is never generated.

To summarise, the diagnostics developed converge on a single conclusion: Sinitic coordinatives are formed from bare acategorial roots merged prior to categorization.

Diagnostics	**Root Coordinatives**	**Lexicalized Compounds**	**aP Parataxis**
Example	*tian-tian-xian-xian*	*tian-suan* (-jiang)	*tian xian (la) de*
Reversibility	✓	× (Fixed)	✓
Strict Binarity	✓	✓(Fixed)	× (Open-ended)
Internal Modification	× (**hen-tian...*)	×	✓(*hen-tian...*)
Independent Redup.	×	×	✓(AA-BB-CC)
Phonological Size	✓(Monosyllabic)	✓(Fixed)	×
Semantic Selection	✓	—	×
M-Word Status	Single M-Word	Single M-Word	≥1 M-Word(s)
Underlying Structure	$\{\sqrt{A}, \sqrt{B}\}$	Stored Atom	[[$_{a\mathrm{P}}$] [$_{a\mathrm{P}}$] …]

Table 1: Structural Diagnostics Separating Root Symmetry from Surface Look-alikes

3 Choiceless Computation and the Legibility of Symmetric Structure

Building on §2, this section must show two things: that deterministic evaluation over a root-symmetric domain forbids selective application, and that the attested outcomes follow from independently motivated interface conditions without construction-specific stipulations. The first is a mathematical result; the second is an empirical claim developed here and in §4.

3.1 Root-Level Set-Merge

Under the SMT, the most economical structure-building operation is Set-Merge, which applies freely without look-ahead (Chomsky 1995; Epstein et al. 2017). When applied to two acategorial, featureless roots, it generates the unordered symmetric set in (13a),

which is subsequently categorized by a functional head (13b).

(13) a. Step 1: Root-level Set-Merge

$\{ \sqrt{A}, \sqrt{B} \}$

b. Step 2: Upward Categorization

$\{_{\text{HP}}\ \text{H}\ \{ \sqrt{A}, \sqrt{B} \}\}$ (where H = a^0, v^0)

While external categorization provides a label for the larger structure via Minimal Search, it does not resolve the internal configuration of the root complex.[4] The domain $\{\sqrt{A}, \sqrt{B}\}$ remains fully symmetric, lacking both hierarchical asymmetry and feature-based distinction. The legitimacy of bare symmetric root-merge in Sinitic is independently established by Zhang (2007) and Hu & Perry (2018), who argue that Sinitic lexicalized compounds arise from merging two categorizer-free roots. The present framework accepts this generative premise and asks a logically subsequent question: what are the algorithmic consequences of such symmetric configurations at the interfaces? While Ott (2015) and Narita & Fukui (2022) have independently pursued symmetry-driven derivational problems, the present framework derives the computational consequences of symmetry as a theorem of deterministic computation rather than constraining generation or positing instability-driven repair.

3.2 Why *AB?

Having established that $\{\sqrt{A}, \sqrt{B}\}$ is a well-formed output of Set-Merge, I now argue that its deviance follows from a fundamental architectural limitation. The symmetric set fails to provide the minimal structural distinction required for deterministic algorithms to operate differentially over its members.

[4] How the categorized structure in (13b) receives its label despite the internal labeling failure of the root complex is addressed in §3.2.2.

3.2.1 God Doesn't Play Dice with the C_{HL}

I distinguish two types of grammatical operations: *constructors* and *evaluators*. The core constructor, $Merge(\alpha, \beta) = \{\alpha, \beta\}$, generates syntactic objects. The freedom required for a finite system to achieve discrete infinity lies entirely in its *unrestricted selection of arguments* (α and β) from the workspace.[5] Evaluators, conversely, do not select inputs to generate new objects. Instead, algorithms like Minimal Search, linearization under LCA, and compositional rules map fixed, already-constructed syntactic domains to legible outputs. As evaluators operate over static structural configurations and lack the generative freedom of argument selection, they are strictly bound by their input. When the input is symmetric, the output must be too.

(14) An algorithm $\mathcal{A}$ is deterministic iff for any well-defined input I, $\mathcal{A}(I)$ yields a unique result: either a well-formed output O (which may be a set, including the empty set $\emptyset$), or failure $\perp$.[6]

Evaluator Determinism is a property already implicit in standard formalizations of grammatical computation. In context-free grammars, each rewrite step is deterministic; non-determinism enters only through rule selection (Chomsky 1956, 1957). In Minimalist Grammars (Stabler 1997), Merge and Move are deterministic over their inputs; optionality arises from the choice of which operation to apply. The same holds for algebraic formalizations of syntactic computation (Marcolli et al. 2025). Across these frameworks, individual mapping steps are deterministic; what varies across derivations is which step is selected. The present framework isolates this property on the evaluator side of the architecture and draws out its consequence for symmetric inputs.

[5] IM is not itself an evaluator as its target is selected by Agree, and it is the deterministic character of Agree that propagates through the CCT. Note also that Parallel Agree (Béjar & Rezac 2009) and Multiple Agree (Hiraiwa 2001), which simultaneously match all goals bearing the relevant features, deterministically return a *set* of matched goals, a unique output, still properly deterministic under Evaluator Determinism.

[6] Throughout this paper, $\emptyset$ denotes a well-formed mathematical empty set (of features or ordering constraints), whereas $\perp$ denotes an undefined or ill-formed algorithmic output (a computation crash).

A second, independent motivation comes from the Inclusiveness Condition. Selective targeting within a structurally symmetric domain requires a choice device, such as Hilbert's ε-operator or a stochastic tie-breaking procedure (cf. Blass et al. 1999). Any such device introduces a derivation-internal distinction between two elements that the syntax itself did not distinguish. That is precisely what Inclusiveness forbids. The simplest evaluator architecture, and the one most consistent with the SMT, is therefore one that eschews arbitrary choice.

(15) **Evaluator Determinism:**

All *evaluators* internal to C_{HL}, and to its external components (PF, LF), are deterministic functions lacking arbitrary choice.[7]

To make this question formally tractable, I characterize structural indistinguishability (17) in terms of automorphisms (16).

A syntactic domain is a tuple $\mathfrak{D} = (D, \sqsubseteq, C, F)$, where D is a finite set of syntactic nodes; $\sqsubseteq$ is the dominance relation; $C \subseteq D \times D$ is the c-command relation derived from $\sqsubseteq$; and $F : D \to \mathcal{P}(\text{Features})$ is a feature-assignment function mapping each node to its set of formal morphosyntactic features. For bare roots, $F(\sqrt{A}) = F(\sqrt{B}) = \emptyset$. What counts as a "feature" for a given evaluator is defined by that evaluator's accessible signature. For Minimal Search and linearization, accessible features are formal morphosyntactic features; encyclopedic indices, by contrast, are accessible to the Encyclopedia but not to MS or the linearization algorithm. This relativization is standard in finite model theory (Blass et al. 1999) and follows from the modularity of the grammar.

(16) **Automorphism:**

An automorphism of a syntactic domain $\mathfrak{D} = (D, \sqsubseteq, C, F)$ with respect to an algorithm $\mathcal{A}$ is a bijection $\sigma : D \to D$ that preserves (i) dominance ($\alpha \sqsubseteq \beta \Leftrightarrow$

[7] Although Merge is also deterministic, the generative freedom of syntax stems entirely from the *selection of arguments* from the workspace, a process that precedes the constructor operation and thus falls outside the scope of the current formulation.

$\sigma(\alpha) \sqsubseteq \sigma(\beta)$), (ii) c-command ($\alpha\, C\, \beta \Leftrightarrow \sigma(\alpha)\, C\, \sigma(\beta)$), and (iii) the feature assignment accessible to $\mathcal{A}$ ($F_{\mathcal{A}}(\sigma(\alpha)) = F_{\mathcal{A}}(\alpha)$ for all $\alpha \in D$).[8] The set of all such automorphisms forms the automorphism group $\mathrm{Aut}_{\mathcal{A}}(\mathfrak{D})$ under composition.[9]

(17) **Indistinguishability:**

Given a syntactic domain $\mathfrak{D}$ with $\alpha, \beta \in D$, $\alpha \approx_{\mathcal{A}} \beta$ iff there exists a permutation $\sigma \in \mathrm{Aut}_{\mathcal{A}}(\mathfrak{D})$ such that $\sigma(\alpha) = \beta$ and $\sigma(\beta) = \alpha$.

Two elements are indistinguishable with respect to an algorithm $\mathcal{A}$ when there exists a structure-preserving permutation that swaps them. Note that indistinguishability is algorithmic equivalence, not ontological identity. $\sqrt{A}$ and $\sqrt{B}$ remain distinct workspace vertices with distinct encyclopedic indices, but their formal profile under $\mathcal{A}$ is invariant under transposition. Considering the minimal syntactic domain containing the symmetric set $\{\sqrt{A}, \sqrt{B}\}$, the transposition $\sigma = (\sqrt{A}\ \sqrt{B})$ preserves all three conditions: mutual c-command, no dominance relation between sisters, and both roots bear empty formal feature sets ($F_{\mathcal{A}} \circ \sigma = F_{\mathcal{A}}$). Therefore $\mathrm{Aut}_{\mathcal{A}}(\{\sqrt{A}, \sqrt{B}\}) \supseteq \{id, \sigma\}$ for every relevant $\mathcal{A}$. As indistinguishability is *algorithm-relative*, the same pair $\{\sqrt{A}, \sqrt{B}\}$ generates opposite verdicts under different evaluators, as in (18a)–(18b).

(18) a. $\sqrt{A} \approx_{\mathrm{MS}} \sqrt{B}$: indistinguishable under Minimal Search (both featureless, mutual c-command; $F_{\mathrm{MS}} \circ \sigma = F_{\mathrm{MS}}$ trivially).

b. $\sqrt{A} \not\approx_{\mathrm{Enc}} \sqrt{B}$: distinguishable under the Encyclopedia ($\sqrt{A}$ maps to $\mathrm{Concept}_A$, $\sqrt{B}$ maps to $\mathrm{Concept}_B$; $F_{\mathrm{Enc}} \circ \sigma \neq F_{\mathrm{Enc}}$).

This algorithm-relativization permits convergence via encyclopedic retrieval in cases where MS-level symmetry would otherwise force crash (§4.3.2).

[8] The evaluator-relativization of Aut in clause (iii) follows standard practice in finite model theory and descriptive complexity (Blass et al. 1999), where automorphisms are routinely relativized to the logical signature accessible to a given computational device.

[9] The algebraic vocabulary here provides the precise starting point for subsequent deduction that informal appeals to 'symmetry' cannot supply. Readers unfamiliar with the algebraic concepts may consult the Appendix for an informal introduction.

Throughout this paper, an automorphism group containing only the identity mapping ($\text{Aut}_{\mathcal{A}} = \{id\}$) is termed *trivial*, denoting a structurally asymmetric domain where every element is uniquely identifiable. A group containing structure-preserving permutations ($\text{Aut}_{\mathcal{A}} \supseteq \{id, \sigma\}$) is *non-trivial*, formalizing unresolved structural symmetry.

Given Evaluator Determinism and the notion of indistinguishability just defined, the following is a mathematical consequence:

(19) **Choiceless Computation Theorem (CCT):**

For any deterministic algorithm $\mathcal{A}$ and syntactic domain $\mathfrak{D}$, if elements $\alpha, \beta \in D$ are indistinguishable with respect to $\mathcal{A}$ ($\alpha \approx_{\mathcal{A}} \beta$), then $\mathcal{A}$ cannot selectively apply to α over β or β over α. **Any well-defined output of $\mathcal{A}$ must treat α and β uniformly**.

Proof. Let $\sigma = (\alpha\, \beta) \in \text{Aut}_{\mathcal{A}}(\mathfrak{D})$. Since σ preserves all properties accessible to $\mathcal{A}$, $\mathcal{A}(\mathfrak{D}) = \mathcal{A}(\sigma(\mathfrak{D}))$. Let $O = \mathcal{A}(\mathfrak{D})$. Suppose $\mathcal{A}$ assigns distinct values to α and β: $O(\alpha) = x \neq y = O(\beta)$, where $O(\gamma)$ denotes the output $\mathcal{A}$ assigns to γ. Then $O(\sigma(\alpha)) = O(\beta) = y$ and $O(\sigma(\beta)) = O(\alpha) = x$, so $\mathcal{A}(\sigma(\mathfrak{D})) \neq \mathcal{A}(\mathfrak{D})$. Contradiction. □

The CCT requires uniform treatment, but it does not itself determine which uniform treatment obtains. Distribution, collapse, and crash are all compatible with the theorem in principle; the choice among them is made only once a series of independently motivated interface conditions are introduced, spelled out in the derivations below. The contribution of the present framework lies not in any one such condition in isolation, but in showing that their interaction is regulated by a single algebraic restriction on deterministic evaluation. Mathematically, the result is a standard consequence of the equivariance of deterministic functions under input automorphisms (Blass et al. 1999; Rossman 2010: 574, Lemma 2); its novelty lies not in the mathematics but in its application to the grammar.[10]

[10]The relevance of (in)distinguishability to syntactic computation has also been independently recognized

A limit case obtains when no invariant is accessible to $\mathcal{A}$. The fixed-point set is empty and $\mathcal{A}$ returns $\emptyset$, as for two featureless roots. The CCT and its limit case jointly underwrite the architectural analysis developed in the remainder of this paper. The scope of Evaluator Determinism and the CCT is summarized in Table 2.[11]

Category	**Role**	**Eval. Det. / CCT**	**Operations**
Constructor	Generates SO from arguments	— / —	Merge (EM, IM)
Evaluator	Maps SO to output	✓ / ✓	MS, Agree, Linearization, VI, FA, …
Performance	Executes interface output	— / —	S-M articulation, parsing

Table 2: Scope of Evaluator Determinism.

3.2.2 The Null Label as a Fourth Labeling Case

When recast in terms of automorphism evaluation, the case structure of Chomsky's (2013, 2015) labeling theory yields a previously undiscussed limit case. The labeling outcome for any syntactic domain $\mathfrak{D}$ is fully determined by its automorphism group and the resulting fixed-point set, as summarized in Table 3.

Configuration ($\mathfrak{D}$)	**Aut$_{\mathbf{MS}}$ Profile**	**Fixed-Point Set (Invariants)**	**MS Output (Label)**
1. $\{H, XP\}$	Trivial ($\{id\}$)	(Structural asymmetry)	H
2. $\{\langle XP\rangle, YP\}$	Trivial ($\{id\}$)	(Asymmetric c-command)	YP
3. $\{XP_{[F]}, YP_{[F]}\}$	Non-trivial ($\{id, \sigma\}$)	$\{F\}$ (Shared feature)	$\langle F, F\rangle$
4. $\{\sqrt{A}, \sqrt{B}\}$	Non-trivial ($\{id, \sigma\}$)	$\emptyset$ (Featureless)	$\emptyset$ **(Null Label)**

Table 3: Labeling Theory as Automorphism Evaluation under the CCT

For Cases 1–2, structural asymmetry forms a trivial automorphism group. Minimal Search deterministically returns the structurally distinguished element. In Case 3 $\{XP_{[F]}, YP_{[F]}\}$, the transposition $\sigma = (XP\,YP)$ yields a non-trivial automorphism group, but the shared feature F belongs to the fixed-point set. As a deterministic algorithm can only return

by Chomsky (2024), citing Reuland (2014). The CCT is consistent with Chomsky's (2005) third-factor principles and can be understood as a formalization of antioptionality (Kayne 2008).

[11] '—' indicates the category falls outside the scope of Evaluator Determinism.

invariant properties, Minimal Search returns $\langle F, F \rangle$, deriving Chomsky's shared-feature label algebraically.[12]

Case 4 constitutes the mathematical limit. When the fixed-point set is strictly empty, Minimal Search does not fail algorithmically. Rather, it deterministically returns the empty set $\emptyset$ as the Null Label.[13] The labeling failure of Sinitic *AB is exactly this predicted terminal output of the CCT: the algorithm succeeds in identifying the absence of invariants, but the C-I interface crashes because a Null Label provides no instructions for categorical projection and compositional semantics.

Independently of the Sinitic application developed below, the Null Label fills a gap in the logical space of labeling theory. Any root-symmetric configuration, including the lexicalized root compounds independently argued for by Zhang (2007) and Hu & Perry (2018), is predicted to receive a null label under Minimal Search. The automorphism-based recasting of labeling thus yields a new logical case while providing an algebraic underpinning for the existing three.[14]

This result highlights a deeper property of Evaluator Determinism itself, independent of the automorphism machinery. Because a deterministic algorithm can only return what is formally present in the workspace, and a bare root $\sqrt{R}$ lacks formal features ($F(\sqrt{R}) = \emptyset$), any labeling evaluator applied to $\sqrt{R}$ or to a root-complement configuration $\{\sqrt{R}, \text{XP}\}$ fails to yield a categorial label on $\sqrt{R}$'s side. Within the present architecture, a bare root cannot independently project, and must therefore enter the derivation

[12] Formally, $\text{Inv}(G) = \{P : D \to \text{Values} \mid P \circ \sigma = P \text{ for all } \sigma \in G\}$. The underlying principle here is that a deterministic algorithm operating over a domain $\mathfrak{D}$ can only return elements of $\text{Inv}(\text{Aut}_{\mathcal{A}}(\mathfrak{D}))$.

[13] Symmetric labeling schemas (e.g., Citko 2011: ch. 5) close no escape hatch for featureless roots. The intersection of two empty feature sets ($F(\sqrt{A}) \cap F(\sqrt{B})$ where both are $\emptyset$) trivially yields $\emptyset$; the union is likewise $\emptyset$. Thus, no non-null label can be recovered to salvage interface legibility.

[14] Under analyses like Harley (2005, 2014), roots can independently project √P and take complements prior to categorization, creating $\{\sqrt{\textsc{Hammer}}, \text{DP}\}$. Within the present architecture, this configuration deterministically returns a Null Label (Case 4), and the C-I interface receives no instruction for compositional interpretation of the root-complement relation, the same failure that excludes $\{\sqrt{A}, \sqrt{B}\}$. Bare roots therefore cannot project without categorial mediation under the present formulation.

via asymmetric merge with a categorizing head.

Note that the internal labeling failure of $\{\sqrt{A}, \sqrt{B}\}$ does not propagate to the containing structure. Recall (13): when a categorizing head H merges with $\{\sqrt{A}, \sqrt{B}\}$, Minimal Search applies to $\{H, \{\sqrt{A}, \sqrt{B}\}\}$. Regardless of the complement's internal label (here $\emptyset$), the structural asymmetry between a terminal head and its complement renders the automorphism group trivial (Case 1). The categorized structure is successfully labeled H, strictly containing the Null Label within the root complex. Containment of the Null Label does not entail interpretive inertness, however. Under bottom-up composition (Heim & Kratzer 1998), the C-I interface must resolve the internal semantic contribution of $\{\sqrt{A}, \sqrt{B}\}$ before integrating it with H. It is this internal resolution that fails (§3.2.3; see §5.1 for the mechanism that succeeds when sufficient structure is provided).

3.2.3 Extending to External Components

The CCT extends beyond labeling within NS to all deterministic evaluators. From the theorem, a general legibility condition obtains directly.

(20) **Computable Asymmetry:**

An SO $\{\alpha, \beta\}$ is legible to deterministic evaluator $\mathcal{A}$ iff $\alpha \not\approx_{\mathcal{A}} \beta$.

PF and LF instantiate this condition over their respective domains.

(21) Domain-specific instantiations of indistinguishability:

a. *Linearization* (PF): $\alpha \approx_{\text{PF}} \beta$ iff the transposition $(\alpha\ \beta)$ preserves all properties accessible to the linearization algorithm. α and β occupy structurally symmetric positions such that neither asymmetrically c-commands the other.

b. *Composition* (LF): $\alpha \approx_{\text{LF}} \beta$ iff the transposition $(\alpha\ \beta)$ preserves all properties accessible to the compositional rules. The structural representation provides no basis for assigning α and β distinct compositional roles.

Linearization succeeds over a domain D iff $\mathrm{Aut}_{\mathrm{PF}}(D)$ is trivial, and mutual c-command denotes a non-trivial automorphism group. The contradictory precedence statements $\langle \alpha \prec \beta \rangle$ and $\langle \beta \prec \alpha \rangle$ are the *output* symptom, while the non-trivial automorphism is the *input* diagnosis. In Malanoski & Kandybowicz's (2026) terms, mutual c-command generates contradictory ordered pairs in the underlying precedence relation P_u, which cannot be resolved into a well-formed surface relation P_s. Thus, the crash is located within the determinism-governed evaluative component, independently of S-M physics.[15] This failure generalizes to deletion-based linearization repairs (cf. Citko 2011). When the indistinguishable elements are featureless roots, no discarding criterion is available.

The automorphism formulation extends the LCA's predictive reach. The bare LCA diagnoses which configurations crash but does not predict how convergence can be restored. The CCT does both: it predicts that convergent strategies exist and constrains their form (§4). It also generates a cross-modal prediction: in sign languages, distinct articulatory channels may supply channel-based features accessible to $\mathrm{Aut}_{\mathrm{PF}}$, thereby trivializing the automorphism group without reduplication. This predicts simultaneous articulation as a modality-specific convergent strategy (cf. Malanoski & Kandybowicz 2026); within a single articulatory channel, by contrast, the same linearization constraints as in spoken language are expected to apply.

The same diagnosis applies at LF. When $\mathrm{Aut}_{\mathrm{LF}}(\mathfrak{D})$ is non-trivial, no compositional rule can assign the symmetric elements distinct semantic roles, and the indeterminacy of role assignment is the output symptom. Table 4 summarizes how this singular algebraic symmetry induces distinct, fatal indeterminacies at each interface.

[15] This two-level separation matters architecturally: if the crash depended on S-M physics, the formal derivation would have a gap. By locating the crash at P_u's failure to yield a well-formed P_s, the derivation remains within the scope of Evaluator Determinism.

Component	Requirement	Symmetric Root Output	Result
NS	Labeling; structural distinction	$\{\sqrt{A}, \sqrt{B}\}$ freely generated	Transferred as-is
S-M Interface	Linear precedence	Non-trivial $\mathrm{Aut}_{\mathrm{PF}} \rightarrow$ contradictory precedence	Non-linearizable ($P_u \nrightarrow P_s$)
C-I Interface	Stable interpretation	Non-trivial $\mathrm{Aut}_{\mathrm{LF}} \rightarrow$ compositional indeterminacy	Compositional failure
Convergence	Interface satisfaction	Requirements unmet	**Crash (*AB)**

Table 4: Symmetric Indeterminacy across Grammar Components

3.3 Crash-tolerant Syntax

The CCT architecture is compatible only with a crash-tolerant grammar. Under strictly crash-proof models (Frampton & Gutmann 2002; cf. Müller 2011), the grammar prevents the generation of interface-failing structures. Applied here, such approaches would need either look-ahead or construction-specific NS-internal repair. Both approaches add computational complexity beyond what the SMT permits.

A direct consequence of locating determinism exclusively within the evaluative component (§3.2.1) is a strictly crash-*tolerant* architecture. Since Evaluator Determinism restricts evaluators but leaves Merge free, the grammar necessarily permits the generation of structures that are locally unresolvable for the evaluators themselves. NS freely generates structural possibilities without evaluating interpretability, and convergence is determined post-syntactically (cf. Chomsky, Gallego & Ott 2019; Chomsky et al. 2023; Chomsky 2024).[16] The symmetric set $\{\sqrt{A}, \sqrt{B}\}$ is transferred intact to the interfaces, where independently motivated algorithms render it legible. In this sense, NS operates non-teleologically: it blindly generates derivational paths, leaving it entirely to the independent evaluation of the interfaces to filter out failures.

The architectural division is captured by two evaluations of the automorphism group.

[16] As Marcolli et al. (2025: 51) formalize, NS cannot introduce an arbitrary index to mark one root and break symmetry prior to Transfer, as this would violate the No Tampering Condition. The CCT supplies the algebraic consequence: when no such index is available, uniform treatment is the only well-defined output.

$\mathrm{Aut}(\mathfrak{D})$ characterizes the symmetry profile of the domain as generated by NS and transferred to the interfaces; $\mathrm{Aut}(\mathfrak{D}_{\text{eval}})$ characterizes the symmetry profile at the point of interface evaluation, after post-syntactic operations have applied. Convergence reduces to the relationship between the two. NS freely generates domains with non-trivial $\mathrm{Aut}(\mathfrak{D})$; the interfaces require trivial $\mathrm{Aut}(\mathfrak{D}_{\text{eval}})$.

Given a crash-tolerant architecture in which evaluative computation is deterministic and therefore subject to the CCT (19), the following corollary obtains:

(22) **Configurational Convergence:**

A derivation over domain $\mathfrak{D}$ converges configurationally iff for every evaluator $\mathcal{A}$ at every interface, $\mathrm{Aut}_{\mathcal{A}}(\mathfrak{D}_{\text{eval}}) = \{id\}$.

While standard Full Interpretation (Chomsky 1986, 1993, 1995) requires that all interface elements bear legible features (e.g., proper valuation), it lacks a formal mechanism to diagnose *configurational* illegibility, namely crashes arising from symmetric structure rather than from unvalued features. The CCT refines this configurational dimension of Full Interpretation into a precise algebraic condition.[17] Table 5 previews the four possible fates of the symmetric input; the mechanisms underlying each are developed in §4 and §5. The remainder of this paper derives each row.

Pattern	**Aut**$(\mathfrak{D})$	**Strategy**	**Aut**$(\mathfrak{D}_{\textbf{eval}})$	**Outcome**
*AB	Non-trivial ($\{id, \sigma\}$)	—	Non-trivial ($\{id, \sigma\}$)	Crash
AABB	Non-trivial ($\{id, \sigma\}$)	Trivialization	Trivial ($\{id\}$)	Converge (§4.1)
ABAB	Non-trivial ($\{id, \sigma\}$)	Dissolution	Trivial ($\{id\}$)	Converge (§4.2)
Stored AB	Non-trivial ($\{id, \sigma\}$)	Stored asymmetry	Trivial ($\{id\}$)	Converge (§4.3.2)

Table 5: Four fates of the symmetric input $\{\sqrt{A}, \sqrt{B}\}$ under crash-tolerant syntax.

[17] A structure with a trivial automorphism group is one in which every element is uniquely determined by its structural position (cf. the orbit-stabilizer theorem).

4 Externalizing Symmetry

I argue that reduplication in Sinitic adjectival coordinatives does not instantiate a pre-specified phonological template, but emerges from the interaction between structurally symmetric inputs and independently motivated interface operations. While previous research has localized these patterns to different structural levels, distinguishing between category-less roots (AABB) and categorized stems (ABAB) (Melloni & Basciano 2018), a fundamental explanatory gap remains: why these specific linearizations should be deterministically tied to their respective configurations. Current accounts typically resort to *ex nihilo* morphophonological templates (e.g., Zhang 2015; Wang 2022; Cheng 2025) or alignment constraints to derive the surface order (Lee-Kim 2016). By contrast, I propose that since NS does not introduce asymmetry within the root complex, the symmetric configuration is transferred to the interfaces without hierarchical differentiation. The observed AABB and ABAB patterns are deterministic strategies of externalization forced by the need to resolve symmetric input under different derivational timings.

4.1 AABB: Post-Syntactic Externalization

As established in §3.2.3, the symmetric set $\{\sqrt{A}, \sqrt{B}\}$ generates contradictory precedence within a single linearization domain. What is required is not the introduction of precedence, but a reconfiguration of the domain over which linearization applies.

4.1.1 Deriving the Pattern

Adjectival predication in Sinitic involves a non-scalar functional projection InstP (Phenomenological Instantiator), which coerces a timeless, individual-level property into a perceptible, spatio-temporally sensitive stage-level predicate (Liu 2013; Wong 2026). Morphosyntactically, Inst introduces a Red as the Vocabulary Item and licenses direct predication. The present analysis adopts InstP over the prominent DegP analysis of

adjectival reduplication (Zhang 2015; Wang 2022), positioned DegP > InstP > aP.[18]

Under Free-Merge and the established crash-tolerant architecture, this symmetric root pair is a fully licit input to subsequent Merge with a^0. The labeling algorithm runs at Transfer for interface evaluation, not as a gatekeeper on structure building.[19]

(23) Syntactic output (Unordered set):

$\{_{\text{InstP}}$ Inst-RED $\{_{a\text{P}}\ a^0\ \{\sqrt{A}, \sqrt{B}\}\}\}$[20]

As affixal morphemes must satisfy their morphotactic requirements at PF, Inst is locally associated with a host via Morphological Lowering (Embick & Noyer 2001; cf. Lee-Kim 2017), targeting the head of its complement and forming a complex Morphological Word (M-Word). As Lowering targets locally the *head* of its complement, which

[18]Two independent diagnostics constitute direct negative evidence against equating Inst with DegP. First, only stage-level predicates can serve as direct complements to perception verbs, which select for observable eventualities rather than abstract propositions.

(1) Perception verb diagnostic (monosyllabic):

a. *wo kandao ta-de lian honghong-de.*
I see 3SG-POSS face red.RED-DE
'I saw his face in a reddish state.' (direct perception of state)

b. ?? *wo kandao ta-de lian hen hong.*
I see 3SG-POSS face DEG red
'I saw his face very red.' (→ coerced to epistemic reading)

Second, individual-level predicates resist episodic temporal modification. Reduplication enables temporal anchoring for bases that otherwise reject it.

(2) Spatiotemporal modification diagnostic (disyllabic):

a. *ta-de naozi gangcai huhututu-de.*
3SG-POSS brain just.now muddled.RED-DE
'His brain was in a muddled state just now.' (episodic state)

b. ?? *ta-de naozi gangcai hen hutu.*
3SG-POSS brain just.now DEG muddled
'His brain was very muddled just now.' (pragmatically odd: stable trait × episodic modifier)

Moreover, in Cantonese, reduplicated adjectives co-occur with degree adverbs (53).

[19]This forecloses the objection that an unlabeled complement should crash the derivation before Lowering. Under Free-Merge, Merge applies blindly without consulting labels on its inputs, and labeling failure matters only at the interfaces.

[20]I assume the phasehood of categorizer (Marantz 2007) and the Phase Impenetrability Condition 2 (Chomsky 2001). Transfer of the complement of a phase head applies upon introduction of the next phase head. Since Inst merges within the extended projection of a^0, the root complex $\{\sqrt{A}, \sqrt{B}\}$ remains accessible at the point where Inst is introduced.

is a^0, it does *not* reach into the root set $\{\sqrt{A}, \sqrt{B}\}$. This is a direct consequence of the labeling failure established in §3.2.2. As the symmetric set $\{\sqrt{A}, \sqrt{B}\}$ returns a Null Label under Minimal Search (Table 3, Case 4), the root complex has no identifiable X^0 head. Lowering, as a strictly Head-to-Head operation (Embick & Noyer 2001), requires an X^0 target in the complement domain. The absence of such a target halts Lowering at a^0, leaving the symmetric set intact for the subsequent stage of VI.

(24) Lowering(Inst, a^0)

$\{_{\text{InstP}} \ldots \{_{a\text{P}} [_{\text{M-Word}}\ a^0 \oplus \text{Inst-}\textsc{Red}]\ \{\sqrt{A}, \sqrt{B}\}\}\}$[21]

The CCT-induced obstacle arises at VI. Under the affixation approach to reduplication in DM (Haugen 2008, 2011; Deal 2016), Red is a Vocabulary Item whose phonological content is computed by copying from a base within a defined locality domain, its morphological sister. In (24), the sister of the complex head $[a^0 \oplus \text{Inst-}\textsc{Red}]$ is the set $\{\sqrt{A}, \sqrt{B}\}$, and it is at this search step that the CCT bites: as the set is symmetric, Minimal Search cannot return a unique base for Red to copy.

The forced resolution follows from the CCT, the Stray Affix Filter (Lasnik 1981), and the structural requirements of Vocabulary Insertion. The baseline case is monoroot AA reduplication (*tian-tian* 'sweet-sweet'), where Minimal Search trivially returns a unique terminal and Red copies its phonological content. When the same VI mechanism encounters a symmetric set $\{\sqrt{A}, \sqrt{B}\}$ rather than a single root, however, it faces two targets rather than one. The CCT excludes selective application; the Stray Affix Filter prohibits non-application; and portmanteau insertion is unavailable because a symmetric domain $\{\alpha, \beta\}$ lacks the internal hierarchical or linear relations required to satisfy a multi-terminal insertion context.[22] Distributive application—the same algorithm extending

[21] Following Embick (2010: 26), $\oplus$ indicates concatenation within a complex M-Word.

[22] A portmanteau VI is indexed to a structural relation *spanning* multiple terminals. A symmetric domain provides multiple terminals but no head, no complement, no specifier—only two structurally indistinguishable sisters. Portmanteau insertion over such a configuration is therefore architecturally

naturally from one target to two—is therefore the only admissible resolution.

Crucially, this extension is not ad hoc. Distributive exponence is independently attested in correlative coordination (Ancient Greek *-te…-te*, Latin *-que…-que*; Mitrović & Sauerland 2014, 2016) and in Kayardild modal case suffixation (Evans 1995), where an affixal morpheme scoping over a structurally flat nominal domain is realized on every member of that domain (§4.1.2). What the CCT contributes is not the existence of distributive exponence itself, but the algebraic condition under which it becomes forced. This is the specific form taken by CCT-governed uniformity when an affixal head scopes over a symmetric domain.

(25) **Distributive Exponence Corollary (DEC):**

When an affixal head H takes scope over a symmetric domain $D = \{\alpha, \beta\}$ where $\alpha \approx_{\mathcal{A}} \beta$, $\mathcal{V}$ applies exhaustively over D: $\mathcal{V}(D) = \{\mathcal{V}(\alpha), \mathcal{V}(\beta)\}$.

While the DEC expresses the outcome algebraically as a set $\{\mathcal{V}(\alpha), \mathcal{V}(\beta)\}$, PF must instantiate this mathematical object as manipulable morphological entities. PF translates the mathematical uniformity over the set into the generation of parallel morphological nodes, forming independent Subwords that host the distributed exponence.[23] As the sister-search of Red returns the symmetric set $\{\sqrt{A}, \sqrt{B}\}$ rather than a unique base, the homomorphic extension of Vocabulary Insertion over the set systematically splits to two parallel Subwords, each containing a single root and constituting an independent locality domain within which copying is unambiguously defined. This is exactly the base-

incoherent, yielding a falsifiable prediction: no attested portmanteau morpheme should have a purely symmetric insertion context. Resolved agreement morphology, which might superficially resemble portmanteau over symmetric controllers (e.g., Slavic default plural with two singular controllers), instantiates a different CCT outcome: invariant extraction over a non-trivial automorphism group with a non-empty fixed-point set (Table 3, Case 3), not portmanteau insertion.

[23]This translation from an algebraic set to duplicated morphological nodes does not violate the No Tampering Condition, which rather constrains narrow syntactic constructors. At PF, post-syntactic morphological operations systematically alter terminal geometries (e.g., Fission, Lowering, Copying) to satisfy language-specific morphotactic requirements. The DEC merely dictates the algebraic boundary conditions under which such morphological duplication is logically forced.

reduplicant correspondence relation internal to a single Subword formalized by Haugen (2008, 2011).

(26) Subword Doubling and Distributive Exponence:

$$\{ \ldots \{_{aP} [_{\text{M-Word}} a^0 \oplus (_{\text{Subword}_1} \text{Inst-Red}_1 \oplus \sqrt{A}) \oplus (_{\text{Subword}_2} \text{Inst-Red}_2 \oplus \sqrt{B})] \}\}$$

The result is two parallel Subwords within a single M-Word as the M-Word boundary is established by the categorizer a^0 in NS. Post-syntactic operations cannot duplicate a^0 or the phase it defines without violating the No Tampering Condition. Subwords are strictly morphological subdivisions of an existing M-Word created by PF to satisfy distributional requirements that the syntactic M-Word structure cannot resolve. This is independently confirmed by the single shared *de* at the right edge of the coordinative (recall §2.2), which diagnoses a single M-Word rather than two independent ones.

Thus, distributive exponence does not create new syntactic structure. It creates parallel morphological attachment sites for a single morphotactic requirement, instantiating one-to-many exponence. The categorizer a^0 and the phase it defines are established once in NS and are not duplicated. The consequence is that the two roots, once embedded in parallel Subwords, no longer stand in mutual c-command within a single linearization domain. The LCA consequently generates *no ordering statements* between them.

The contrast with a monoroot complement confirms the logic. When a reduplicant takes scope over a single root, as in AA reduplication (*tian-tian* 'sweet-sweet'), Minimal Search trivially returns the unique terminal and Red copies its phonological content.[24] When Red takes scope over the symmetric set $\{\sqrt{A}, \sqrt{B}\}$, both roots are simultaneously returned and neither can be uniquely selected. The same algorithm, applied to inputs with different automorphism profiles, deterministically generates different outcomes.

[24] I assume that Minimal Search also applies at PF as a third-factor efficiency condition on computation, not as a module-specific algorithm restricted to NS. Under this view, any search operation over syntactic structure, whether for labeling at Transfer or for identifying a reduplicative base at VI, instantiates the same minimal computational procedure.

The AABB pattern thus instantiates *obligatory* distributive exponence under structural symmetry. Multiple exponence is independently attested (e.g., Arregi & Nevins 2012; Harris 2017; Malanoski & Kandybowicz 2026) but typically treated as optional or as the result of localized feature mismatch. Here, its obligatoriness follows from architectural principles: when a morphotactic requirement cannot be resolved at a unique terminal, distributive exponence is the *only* admissible outcome.

VI proceeds independently within each Subword. Under cyclic VI (Bobaljik 2000), roots receive their phonological exponents before RED, so by the time RED is evaluated, the root already supplies a phonological base for copying. The domain of phonological copying Φ is therefore localized. RED_1 copies $\sqrt{A}$ and RED_2 copies $\sqrt{B}$, forming the observed AABB pattern.

(27) i) VI($\sqrt{A}$, $\sqrt{B}$) and, ii) Localized Copying

a. Subword 1: $\Phi(\text{RED}_1) \rightarrow \text{A} \oplus \text{A}$

b. Subword 2: $\Phi(\text{RED}_2) \rightarrow \text{B} \oplus \text{B}$

The linear order *within* each Subword is also not determined by the CCT. Within a single Subword containing one root and its reduplicant, there is no residual symmetry. Intra-Subword ordering is a matter of language-specific morphotactics, and the proposal stays agnostic on whether a given language realizes reduplicants as prefixal or suffixal.

In algebraic terms, Subword Doubling functions as a *trivialization* of the automorphism group. $\{\sqrt{A}, \sqrt{B}\}$ constitutes a non-trivial Aut_{PF}; Subword Doubling partitions this domain into independent subdomains each containing a single root to render Aut_{PF} trivial within each. The constraint set *between* subdomains is empty. Thus, under cyclic linearization (Fox & Pesetsky 2005; Richards 2010), both $\langle \text{M-Word}_A \oplus \text{M-Word}_B \rangle$ and $\langle \text{M-Word}_B \oplus \text{M-Word}_A \rangle$ satisfy the grammatical output with zero violations. While the grammar deterministically outputs a single unordered representation with $\emptyset$ inter-Subword ordering constraints, the specific temporal sequence observed in speech is an

epiphenomenal resolution at the S-M interface since the vocal tract cannot articulate two syllables simultaneously in physical time, the performance system must serialize the underspecified morphosyntactic output.[25]

(28) Linear Permutation at PF

a. $[_{\text{M-Word}}\ (_{\text{Subword}}\ \text{A} \oplus \text{A}) \oplus (_{\text{Subword}}\ \text{B} \oplus \text{B})]$

b. $[_{\text{M-Word}}\ (_{\text{Subword}}\ \text{B} \oplus \text{B}) \oplus (_{\text{Subword}}\ \text{A} \oplus \text{A})]$

More interestingly, the AABB pattern is phonologically highly marked as it incurs consecutive violations of the OCP by placing identical syllables and tones in strict adjacency. If phonological optimization were the primary driver, the grammar should *avoid* AABB, not *require* it. The fact that the output preserves the symmetric organization even at the heavy cost of phonological markedness directly supports a view under which PF exponence is constrained primarily by morphosyntactic requirements.

4.1.2 Justifying the DEC

The DEC is not specific to Sinitic reduplication. This subsection shows that distributive exponence recurs whenever an affixal functional head scopes over a symmetric complement, and that the CCT prohibits single-point exponence in each case.

(29) a. *you-ke-ai-*(you)-piao-liang*
CORR-cute-*(CORR)-beautiful
'both cute and beautiful'

b. *bu-nan-*(bu)-nv*
not-man-*(not)-woman
'neither male nor female' (androgynous)

c. *yi-zhang-*(yi)-chi*
one-tight-*(one)-loose

[25] Unlike the cross-linguistically variable but language-internally fixed orderings that Embick (2010) derives for specific morpheme classes, both the AABB and ABAB alternations exhibit true surface optionality, in Subword sequencing and root sequencing, respectively. By locating this optionality in the S-M performance requirement for temporal serialization rather than in the narrow grammar, Evaluator Determinism is strictly preserved: the grammar deterministically outputs a single unordered set; the articulator physically serializes it.

'alternating tension and relaxation'

Each pattern instantiates the DEC over a different functional head (*you* 'both', *bu* 'not', *yi* 'one'), supporting that distributive exponence is a property of any affixal morpheme whose morphotactic requirement collides with a symmetric target.

Cross-linguistically, the DEC effect also surfaces in correlative coordination (Ancient Greek *-te…-te*, Latin *-que…-que*; Mitrović & Sauerland 2014, 2016) and in Suffixaufnahme over non-configurational nominal domains. Kayardild modal case suffixation (Evans 1995) especially provides a transparent non-Sinitic instantiation. An affixal case morpheme scoping over a structurally flat nominal domain is distributively realized on *every* nominal. As Kayardild nominals are categorized individually, their indistinguishability is also *relative* to the case-assigning algorithm. This further stabilizes the claim that the DEC applies whenever $\alpha \approx_{\mathcal{A}} \beta$ with respect *to the relevant algorithm*. See also Warlpiri case stacking (Hale 1982).

4.2 ABAB: Cyclic Spell-Out and Early Linearization

If the AABB pattern reflects post-syntactic morphotactic operations triggered by Lowering, a central question arises: how does the grammar derive the ABAB alternation from the same symmetric root complex $\{\sqrt{A}, \sqrt{B}\}$? As noted in §2, ABAB forms systematically encode a dynamic or iterative construal (Sui & Hu 2016, 2021), contrasting with the stative interpretation associated with AABB. This semantic bifurcation is the first indication that the two patterns are not phonological variants of a single derivation, but reflexes of structurally distinct configurations established prior to Transfer.

Contrary to the derivation of the AABB pattern, ABAB reduplication is built on a distinct syntactic spine. Following Sui & Hu (2021; see also Arcodia et al. 2014), the bases that license ABAB behave as change-of-state predicates. Thus, rather than a stative categorizer a^0, the symmetric root complex merges with a verbalizer v^0, and the relevant

reduplicative morphology is introduced by a higher aspectual head Asp encoding delimitative or tentative aspect (Sui & Hu 2022).

(30) $\{_{\text{AspP}}$ Asp-RED $\{_{v\text{P}}\ v^0\ \{\sqrt{A}, \sqrt{B}\}\ \}\}$

Unlike the adjectival configuration, v^0 and Asp in Sinitic participate in Head-movement. Following Harizanov & Gribanova (2019), I assume that the displacement is a genuinely syntactic operation occurring within NS, while its morphotactic effects are evaluated at PF. The V-to-Asp step is generally motivated in the Mandarin literature (Tang 1998; Paul 2015). Sui & Hu (2016) provide direct diagnostics via the A-not-A intervention effect.

(31) a. *ni kan-bu-kan zhe-ben shu*?
you read-NEG-read this-CL book
'Will you read this book?'

b. * *ni kan-kan-bu-kan-kan zhe-ben shu*?
you read-read-NEG-read-read this-CL book
(Intended: 'Will you do some reading of this book?')

When v^0 probes its complement, Minimal Search encounters the symmetric set $\{\sqrt{A}, \sqrt{B}\}$. Standard Head-movement seeks to target a unique head (X^0) within its complement. However, what symmetry determines here is not *whether* movement applies, but *what* it can target. When the affixal probe v^0 seeks a unique root dependency, the CCT dictates that selective extraction of a proper sub-part (either root) from the symmetric set is mathematically undefined. Under the simplest formulation of Merge (Chomsky 2020, 2024; Chomsky et al. 2023), syntactic operations target SOs generally. This strips away the theoretical primitive of 'X^0 vs. XP' movement. Given the No-Tampering Condition (Chomsky 2008), the probe must therefore attract the minimal containing SO that it can uniquely identify, triggering the exhaustive pied-piping of the entire root set into v^0.[26]

[26]This is fully compatible with the M-Merger framework of Matushansky (2006), where the morphological fusion step applies to the output of syntactic displacement irrespective of the internal complexity of the attracted element.

Crucially, crash is not a viable alternative to pied-piping. Because v^0 bears an affixal requirement, failure to attract any element leaves v^0 unhosted, violating the Stray Affix Filter at PF. Since crash is the only alternative to pied-piping, and since pied-piping of the minimal containing SO is well-defined under the No-Tampering Condition, pied-piping is the sole convergent derivation.

The result parallels Cable's (2010) generalization that when a probe cannot uniquely identify a subpart of its target domain, it attracts the minimal containing SO, as in wh-pied-piping. The present case differs in one important respect. In Cable's system, pied-piping is triggered by intervention: the probe cannot reach past the containing DP shell. Here, by contrast, the probe reaches both candidates but cannot distinguish between them. The trigger is therefore not intervention but indistinguishability. The shared deeper generalization is that whenever a probe cannot return a unique target, it attracts the minimal containing SO. The CCT supplies the principled algebraic trigger for the indistinguishability case: subpart selection is undefined precisely when the automorphism group of the domain is non-trivial.

The same algebraic logic extends naturally beyond the X^0 domain. If a probe confronts a symmetric domain whose members are indistinguishable with respect to the relevant evaluator, selective extraction is undefined under the CCT. The only convergent options are therefore exhaustive extraction or no extraction at all. I return in §4.4.1 to the phrasal reflex of this restriction in ATB extraction.

In the present case, the restriction is instantiated as pied-piping of the root set under v^0:

(32) Root-to-v^0 Pied-piping under symmetry:

$\{_{\text{AspP}}$ Asp-Red $\{_{v\text{P}}\ [_v\ v^0 \oplus \{\sqrt{A}, \sqrt{B}\}]\ \ldots\langle\ \{\sqrt{A}, \sqrt{B}\}\ \rangle\ \}\}$

Once the complex head is formed via this symmetric pied-piping, it behaves as a single syntactic atom for subsequent operations. When the higher aspectual head Asp probes

its complement, it targets this newly formed complex v^0 head and attracts it, culminating in the standard Head-movement of the entire v-complex into Asp:

(33) v-to-Asp Head-movement:

$\{_{\text{AspP}}\ [_{\text{M-Word}}\ \textsc{Asp-Red} \oplus [_v\ v^0 \oplus \{\sqrt{A}, \sqrt{B}\}]]\ \ldots\langle v\rangle\ \ldots\}$

This operation does not resolve the internal symmetry of the root complex. The central effect lies in the creation of a single, bounded cyclic domain prior to Transfer. Under Harizanov & Gribanova (2019), syntactic Head-movement produces a morphologically unified M-Word whose internal organization is subsequently governed by morphotactic rather than narrow-syntactic principles (Embick 2010; Arregi & Pietraszko 2021).

The dissolution strategy does not by itself guarantee convergence. Without Inst or Asp within the extended projection of the categorizer, no functional head triggers Lowering or Head-movement into the root domain prior to Transfer. As roots cannot bypass categorization (Embick & Marantz 2008; cf. §3.2.2), the root complex is invariably spelled out *in situ* at the aP/vP phase boundary, where the LCA encounters mutual c-command and returns $\perp$. Functional heads introduced above the phase boundary (e.g., Pred, T, C) are structurally prevented from accessing the root set by the Phase Impenetrability Condition (Chomsky 2001). The dissolution path is therefore architecturally restricted. It is available exclusively when the extended projection of a categorizer independently provides a head bearing an affixal reduplicative requirement prior to phase closure.[27]

M-Word-internal linearization is not governed by the LCA but by language-specific morphotactic constraints (Baker 1985; Embick 2010). The contrast with AABB reduces to derivational timing: in ABAB, Head-movement incorporates $\{\sqrt{A}, \sqrt{B}\}$ into v^0 *prior*

[27]This restriction also follows from the automorphism formulation of Full Interpretation (22): even if a hypothetical functional head above the phase boundary could somehow access the root set and render PF convergent via dissolution, the C-I interface independently requires a predicational domain established by Inst or Asp for Default PM (§5.1). Without such structure, interpretive instability persists and the derivation crashes at C-I. The *AB ban is thus doubly enforced: by the PIC at PF, and by the absence of a predicational domain at C-I.

to Transfer, rendering the internal geometry opaque to the LCA. In AABB, Lowering is *post-syntactic*, the roots remain independent terminals at Transfer, and the LCA encounters the symmetric domain.

This does not mean the grammar optionally generates two distinct linearizations. The morphotactic component, functioning as a deterministic evaluator, searches for language-specific template constraints on the relative order of the twin roots. When no such template dictates their order, the algorithm outputs an empty set of ordering statements ($\emptyset$). Under Cyclic Linearization (Fox & Pesetsky 2005), convergence requires the absence of contradictory statements ($\perp$), not the presence of exhaustive ordering. As established for AABB (§4.1.1; fn. 25), the specific temporal order observed in speech is not a grammatical choice but an epiphenomenal S-M resolution of the $\emptyset$ morphosyntactic output.

Once linearization is established, the reduplicative operation introduced by Asp applies over the entire M-Word. Since Head-movement fuses the elements prior to VI (Matushansky 2006; Keine & Bhatt 2016), reduplication targets the entire stem:

(34) Cyclic linearization

a. VI($\sqrt{A}$, $\sqrt{B}$): [$_{\text{M-Word}}$ Asp-Red $\oplus$ ($_{\text{Subword}}$ A $\oplus$ B)]

b. Copying: [$_{\text{M-Word}}$ ($_{\text{Subword}}$ A $\oplus$ B) $\oplus$ ($_{\text{Subword}}$ A $\oplus$ B)]

Thus, the ABAB derivation resolves the non-trivial automorphism group by *dissolution* rather than trivialization. Head-movement incorporates the root complex into v^0. The resulting M-Word is linearized by morphotactics, but no longer the LCA. The automorphism group that was non-trivial with respect to LCA linearization is not rendered trivial, but the computational regime shifts entirely.

I formalize the two strategies as follows.

(35) **Trivialization and Dissolution.** Let $\mathcal{A}$ be a deterministic algorithm operating over a domain D with $\text{Aut}_{\mathcal{A}}(D) \neq \{id\}$. Two strategies are available, each licensed by distinct derivational configurations.

a. **Trivialization:** D is decomposed into a family of subdomains $\{D_1, \ldots, D_n\}$ such that $\mathrm{Aut}_{\mathcal{A}}(D_i) = \{id\}$ for each D_i; $\mathcal{A}$ then applies independently within each subdomain.

b. **Dissolution:** D is transferred to a distinct computational regime governed by an algorithm $\mathcal{A}'$ with respect to which $\mathrm{Aut}_{\mathcal{A}'}(D)$ is not evaluated, so that the symmetry relevant to $\mathcal{A}$ never enters $\mathcal{A}'$'s computation.

One might ask why M-Word-internal $\emptyset$ yields underspecification while cross-M-Word $\perp$ yields crash. Contradiction is not absence. The LCA *actively derives* precedence from c-command. Mutual c-command produces $\langle A \prec B \rangle \wedge \langle B \prec A \rangle$, violating the asymmetry condition on strict total orders, inducing $\perp$. Morphotactics searches for language-specific ordering templates. When no template matches, the output is $\emptyset$, a well-formed empty set, not a contradiction. It is the S-M performance system that maps this well-formed underspecification onto a specific temporal sequence.

The derivational bifurcation between AABB and ABAB constitutes an empirical argument for the architectural distinction between NS-internal Head-movement and post-syntactic Lowering. Under the Generalized Head Movement (GenHM) framework (Arregi & Pietraszko 2021), both operations are unified as syntactic displacement. This eliminates the derivational timing difference that the present analysis exploits. If both applied within the same narrow-syntactic regime, the root complex $\{\sqrt{A}, \sqrt{B}\}$ would invariably face the CCT under conditions where only syntactic pied-piping is available, incorrectly predicting uniform stem-level reduplication (ABAB) across both configurations.[28] The Sinitic data therefore provide direct evidence that Lowering must remain post-syntactic to preserve the modular distinction on which the trivialization/dissolution contrast de-

[28]The incompatibility is specific to AABB, where the post-syntactic status of Lowering is load-bearing: it ensures the symmetric root set enters the morphological component, triggering the CCT under conditions where distributive exponence (rather than syntactic pied-piping) is the only available repair.

pends.[29] Table 6 summarizes the derivational bifurcation.

Property	**AABB**	**ABAB**
Categorizer	a^0 + Inst	v^0 + Asp
Derivational Timing	Post-syntactic (Lowering)	NS (Head-movement)
Linearization Domain	Multiple M-Words	Single M-Word
Resolution Strategy	Trivialization of Aut_{PF}	Dissolution of Aut_{PF}
Reduplication Scope	Root-level (AA-BB)	Stem-level (AB-AB)
Interpretation	Stative	Eventive

Table 6: Derivational Profiles of AABB and ABAB Patterns

4.3 On the Licensing of Reduplication

This subsection addresses two licensing questions. §4.3.1 asks why symmetric Set-Merge does not freely overgenerate bare *AB outputs. §4.3.2 asks how the grammar nonetheless converges on lexicalized AB compounds whose PF step should fail.

4.3.1 Licensing and the *AB Gap

Distributive exponence is not a freely available repair strategy for unlinearizable symmetry. It is strictly parasitic on a local morphotactic requirement—the Stray Affix Filter. Affixal heads such as Inst must attach to an overt host. Free functional items such as temporal adverbs bear no such requirement. If a functional head lacking an affixal requirement merges with $\{\sqrt{A}, \sqrt{B}\}$, Vocabulary Insertion does not attempt to distribute it. Consequently, the domain is not restructured, the LCA encounters the unresolved symmetric roots, and the derivation crashes at PF. Doubling is derivationally forced only when a local affixal requirement collides with an indistinguishable target.

Importantly, Inst is independently motivated. Inst is a regular functional head in the Sinitic adjectival domain responsible for reduplication across multiple patterns (Wong 2026), as illustrated in (36a–36b). In monoroot cases, Vocabulary Insertion trivially iden-

[29] While recent minimalist inquiries leave open the possibility that all head displacement might be relegated to post-syntactic externalization (Chomsky et al. 2023: 66), the Sinitic derivational bifurcation presents direct empirical evidence to the contrary. This empirically supports a mixed architecture (Harizanov & Gribanova 2019) over a uniform post-syntactic treatment.

tifies a unique target and RED attaches without triggering distributive exponence. It is only when Inst takes scope over a symmetric complement that distributive exponence becomes the sole convergent outcome.

(36) a. Partial adjectival reduplication:
nadou ci-qilai nian-hu-hu de
Natto eat-INCH sticky-IDEO-IDEO DE
'Natto tastes sticky.'

b. Disyllabic compound reduplication:
zhe-ge ren qi-qi-guai-guai de
this-CL person strange-strange DE
'This person is strange.'

In sum, the distribution of AABB and ABAB follows from the interaction between symmetric structure and independently motivated interface conditions.

4.3.2 Revisiting the Lexical Escape Hatch: Three Routes

The *AB gap delimits the Syntactic route from below: whenever symmetric Set-Merge applies to $\{\sqrt{A}, \sqrt{B}\}$ and no affixal head is locally present, the LCA returns $\perp$ at PF (§3.3). A complementary question concerns simplex lexicalized AB compounds like *ganjing* 'clean' or *suan-tian* 'sour-sweet', which are also root-merged (Zhang 2007; Hu & Perry 2018) but appear to escape this constraint (recall §2.3.1). I argue that convergence is delivered by the post-syntactic Encyclopedia, acting as an interface-bridging list (Harley & Noyer 2000; Borer 2013; Hu & Perry 2018; Everaert 2023) that rescues the PF configurations the CCT predicts to fail. A symmetric root pair may be accessible through up to three derivational routes, all built on the same Set-Merge configuration:

(37) THREE ROUTES OVER SYMMETRIC ROOT PAIRS $\{\sqrt{A}, \sqrt{B}\}$:

a. **Lexical Route (L)**: Categorized by a^0/v^0 without higher functional heads bearing affixal requirements (e.g., Inst, Asp). At the a/vP phase boundary, the symmetric set $\{\sqrt{A}, \sqrt{B}\}$ undergoes parallel interface evaluation. At C-I, en(cyclopedia)-search retrieves a stored Listeme mapping the set to a holistic

conceptual Gestalt. At PF, standard LCA evaluation fails due to mutual c-command. However, convergence is rescued by a stored lexically specific precedence statement (e.g., $\sqrt{A} \prec \sqrt{B}$).[30] This idiosyncratic morphotactic instruction overrides the general LCA failure, licensing the simplex AB form.[31]

b. **Syntactic Route S_1 (v^0-level)**: Categorized by v^0 + Asp; Head-movement derives the ABAB form with a change-of-state reading.

c. **Syntactic Route S_2 (a^0-level)**: Categorized by a^0 + Inst; Subword Doubling yields AABB with a stative reading.

The availability of each route is jointly determined by two independent conditions evaluated in sequence: (i) whether the Encyclopedia contains a stored Listeme indexed to the root pair, licensing Route L; and if not, (ii) whether the *Dimensionality Constraint* (DC; developed in §5.3) is satisfied—whether the encyclopedic sortal categories of the two roots overlap (e.g., both denoting taste), permitting a C-I-convergent conjunctive parse and thus licensing the compositional readings on S_1/S_2. Table 7 illustrates the partition across three diagnostic root pairs.

	ganjing 'clean'	***suan-tian*** 'sour-sweet'	***tian-xian*** 'sweet-salty'
Encyclopedia entry			
Route L (simplex AB)	✓	✓	×
DC satisfied			
Route S_1 (ABAB)	× (dry ∩ purified)	✓	✓
Route S_2 (AABB)	× (dry ∩ purified)	✓	✓

Table 7: Route availability under the $\{\sqrt{A}, \sqrt{B}\}$ configuration

[30]The capacity for PF to impose item-specific concatenation rules on designated terminals is formally established in Embick (2010).

[31]The imposition of linear order at PF does not fuse the roots into a single structural atom (cf. De Belder 2017; Everaert 2023); the underlying symmetric geometry of the set $\{\sqrt{A}, \sqrt{B}\}$ remains intact. Thus, if a reduplicative head scopes over this lexicalized compound, the CCT strictly applies: selective targeting remains undefined, forcing Subword Doubling (DEC) and yielding the obligatorily distributive ABAB pattern, rather than the AABB pattern expected of true disyllabic monoroots.

Route L does not violate Y-model modularity. En-search (Borer 2013) applies at phase closure, architecturally prior to interface-specific evaluation. The stored Listeme supplies both a conceptual Gestalt for C-I and a precedence statement for PF. Once en-search succeeds, these stored instructions pre-empt interface defaults: the LCA, operating post-Transfer, finds its required ordering already supplied and does not re-derive it from the bare symmetric set. Route L therefore supports the CCT's counterfactual prediction: the LCA *would* return $\perp$ on the bare symmetric set, but en-search preemptively supplies asymmetric instructions before the crash point.

4.4 The 2×2 Typology of Exhaustive Operations

The derivations developed so far support a broader generalization. AABB and ABAB are not isolated construction-specific outputs, but two cells in a larger typology of exhaustive operations over symmetric domains. In each case, the grammar confronts a domain over which the CCT excludes selective targeting; convergence is possible only if the operation applies uniformly. What varies is not whether the CCT applies, but when evaluation takes place and whether the symmetric elements are distinct or identical with respect to the relevant algorithm.

ATB extraction and RNR are used here as canonical calibration cases rather than exhaustive reanalyses of either literature. Their role is to show that the CCT's reach extends beyond the Sinitic discovery domain and that the typology is not ad hoc.

The crucial variable distinguishing distribution from convergence is the identity profile of the symmetric elements. When the relevant elements are distinct, uniform treatment entails independent application; when they are identical, uniform treatment entails no basis for maintaining separate externalization events. Combined with the derivational timing distinction already established for AABB and ABAB, this yields the following four-cell partition.[32]

[32] $\equiv$ and $\not\equiv$ denote identity of phonological or featural content, not workspace token identity. ATB targets

	Distinct exponents ($\alpha \not\equiv \beta$)	**Identical exponents** ($\alpha \equiv \beta$)
Syntax: Dissolution (narrow-syntactic)	ABAB exhaustive pied-piping	ATB exhaustive extraction
PF: Trivialization (post-syntactic)	AABB distributive exponence	RNR convergent deletion

Table 8: 2×2 typology of exhaustive operations over symmetric domains.

The completeness of this partition can be stated as a corollary of the CCT:

(38) Let $\mathcal{A}$ be a deterministic algorithm operating over a symmetric domain D with $\mathrm{Aut}_{\mathcal{A}}(D) \neq \{id\}$. The form of the convergent output is determined by two parameters: (i) *algorithm timing* (narrow-syntactic vs. post-syntactic), which determines the resolution strategy (dissolution vs. trivialization); and (ii) *identity profile* ($\alpha \equiv \beta$ vs. $\alpha \not\equiv \beta$), which determines whether uniform treatment yields convergence (identical elements) or distribution (distinct elements). The resulting 2×2 partition is mathematically exhaustive. A symmetric resolution strategy falling outside these four cells would falsify the domain-generality of the CCT.

Prominent accounts of ATB appeal to Sideward Movement (Nunes 2004) or Multidominance (Citko 2005); prominent accounts of RNR appeal to rightward movement (Sabbagh 2007) or stipulated uniformity (Belk et al. 2023); prominent accounts of AABB and ABAB invoke construction-specific templates (Zhang 2015; Wang 2022; Cheng 2025). The present account derives all four from Set-Merge, Evaluator Determinism, and independently motivated interface conditions. No additional operation-specific or construction-specific machinery is required.

Each cell is occupied and independently attested, constituting convergent evidence that the CCT captures a genuine architectural property of the grammar. Under the present architecture, selective operations over symmetric domains are excluded.

distinct workspace tokens with indistinguishable feature profiles, while AABB targets tokens with distinct encyclopedic indices.

The present typology targets the maximally clear endpoints of the identity dimension, namely fully distinct and fully identical elements. Cases of partial phonological overlap raise a further question about how much identity the PF evaluator treats as relevant; that question is left for future investigation.

The typology above characterizes operations that restructure the symmetric input: trivialization partitions the domain at PF, while dissolution shifts the domain into a distinct computational regime in syntax. LF possesses neither restructuring mechanism. Compositional rules operate over the transferred domain as such, without post-syntactic geometric manipulation. LF convergence therefore proceeds not by restructuring, but by applying the unique σ-invariant compositional rule available, namely Default PM, developed in §5.

The typology is only useful if its non-Sinitic cells can also be independently filled. I turn first to ATB extraction, which instantiates the syntax-identical cell, before considering RNR, the PF-identical cell.

4.4.1 ATB as Exhaustive Extraction

ATB extraction is used here as a canonical calibration case for the syntax-identical cell. This subsection is not intended as a full reanalysis of the ATB literature, but aims to show that the same restriction that forces root-level pied-piping in ABAB extends directly to phrasal extraction over symmetric conjuncts. Once a probe simultaneously accesses two conjuncts that are indistinguishable with respect to the relevant evaluator, selective extraction is excluded by the CCT. The grammar must therefore either extract exhaustively or not extract at all. ATB is the convergent option when extraction proceeds uniformly across the symmetric domain.

The parallel with ABAB can be stated directly:

(39) Shared derivational logic of ATB and ABAB:

a. Probe $\mathcal{A}$ targets a domain containing $\{\alpha, \beta\}$ where $\alpha \approx_{\mathcal{A}} \beta$

b. Select($\{\alpha, \beta\}$) is undefined (CCT)

c. Resolution: $\mathcal{A}$ applies exhaustively over $\{\alpha, \beta\}$

i. ATB: exhaustive extraction $\rightarrow$ both α, β displaced uniformly

ii. ABAB: exhaustive incorporation $\rightarrow$ both α, β pied-piped into an M-Word

This logic immediately yields the symmetric case of the Coordinate Structure Constraint. For a coordinate structure containing $\{\alpha, \beta\}$ such that $\alpha \approx_{\mathcal{A}} \beta$, selective extraction of either conjunct alone is undefined. The only convergent options are therefore exhaustive extraction or no extraction at all.

(40) For a coordinate structure containing $\{\alpha, \beta\}$ such that $\alpha \approx_{\mathcal{A}} \beta$, $\mathcal{A}$ can extract from $\{\alpha, \beta\}$ only exhaustively (ATB) or not at all.

Nothing in this formulation requires a flat coordination geometry. The set notation $\{\alpha, \beta\}$ denotes evaluator-relative indistinguishability, not a commitment to symmetric phrase structure. Under a standard asymmetric spine [&Pα[&′&β]] (Munn 1993; Kayne 1994), the CCT applies whenever an external probe accesses both conjuncts simultaneously in such a way that they are indistinguishable with respect to the relevant evaluator. What matters is not the global geometry of the coordination phrase, but the accessibility conditions that define the probe's search space.[33]

The residual ban on selective extraction from featurally *distinct* conjuncts does not follow from the CCT. Once $\alpha \not\approx_{\mathcal{A}} \beta$, the automorphism group relevant to extraction is trivial, and the algebraic source of ATB uniformity disappears. The asymmetric component of the CSC therefore requires an independent locality-theoretic explanation (cf. Bošković 2020a; Sportiche 2024). The present claim is correspondingly narrower but more precise: the CCT derives the *symmetric* component of the restriction.

[33]For one way of making this precise, see Ke's (2024) parallel Minimal Search, under which a probe may return a match-set rather than a single token. The present argument does not depend on that implementation in particular; it depends only on the existence of an evaluator-relative notion of simultaneous accessibility.

Bošković's (2020a) *Non-ATB ATB* cases provide further support. In such configurations, extraction must proceed from each conjunct, but the extracted elements are not token-identical:

(41) Which book$_i$ and which magazine$_j$ did John buy t_i and Bill read t_j respectively?

Under the present framework, this is exactly what one expects once the extraction domain is symmetric but the extracted elements remain distinct. Selective extraction from only one conjunct is excluded; exhaustive extraction across the symmetric domain is forced. In this respect, Non-ATB ATB stands to phrasal extraction as ABAB stands to root-level incorporation: both instantiate exhaustive application over a domain whose members are distinguishable in content but inaccessible to selective targeting.[34]

If adjuncts are structurally conjunct-like—for example, if they involve a null coordinating head (Higginbotham 1985; Bošković 2020b)—the same logic plausibly extends to the Adjunct Condition. On such a view, parasitic gap constructions would instantiate yet another reflex of exhaustive extraction over a symmetric domain. I leave that extension programmatic here.

ATB thus fills the syntax-identical cell not by introducing a new mechanism, but by extending the same algebraic logic already visible in ABAB: when selective targeting is undefined over a symmetric domain, extraction must proceed exhaustively or not at all.

4.4.2 Right Node Raising as Convergent Deletion

If ATB calibrates the syntax-identical cell, RNR calibrates its PF-identical counterpart. Here the same CCT-imposed uniformity requirement applies not to extraction, but to deletion. Once the relevant elements are both parallel and phonologically identical, PF must treat them uniformly. The aim is not to adjudicate among structural implementations

[34] Bošković's (2020a) generalization that unlabeled elements do not function as interveners aligns with Computable Asymmetry (20): an unlabeled element is one whose automorphism group is non-trivial, and an element with non-trivial Aut cannot be uniquely identified by the probing algorithm, and hence cannot function as a uniquely specified intervener.

of RNR, but to identify the algebraic condition that any successful implementation must satisfy.

Consider [&P[Conj$_A$ ... SE$_1$][Conj$_B$ ... SE$_2$]], where SE$_1$ and SE$_2$ have identical phonological exponents and occupy parallel conjunct-final positions. Indistinguishability is evaluated here relative to the PF-deletion algorithm's accessible signature: phonological content and position relative to the conjunct edge. For the purposes of PF deletion, the pair $\{\mathrm{SE}_1, \mathrm{SE}_2\}$ therefore forms a non-trivial automorphism group over parallel positions with identical exponents, regardless of the global geometry of the surrounding &P.

(42) a. John likes, and Mary hates, *linguistics*. (RNR: convergent deletion)

b. *John likes *linguistics*, and Mary hates. (selective deletion: blocked)

c. *John likes, and Mary hates *linguistics*. (selective deletion: blocked)

The CCT prohibits selective deletion, directly ruling out (42b–c). What remains is to determine which *uniform* treatment obtains. Three logical possibilities present themselves. First, PF could pronounce both copies independently. This is exactly what happens when $\alpha \not\equiv \beta$, as in the AABB cell. When $\alpha \equiv \beta$, however, PF lacks an independent basis for distinguishing two pronunciation events from one. Second, PF could delete both copies, but this would violate recoverability. Third, PF could collapse the two events into one—convergent deletion. This is the only remaining option.

The distribution/convergence split is therefore not a free parameter but follows from the interaction of CCT uniformity with the identity profile of the symmetric pair. When elements are distinct, distributing and collapsing yield different surface strings; only distribution faithfully externalizes both. When elements are identical, the two strategies are observationally equivalent, and economy favors collapse.

What determines the surface position of the surviving copy? Under Cyclic Linearization (Fox & Pesetsky 2005), each conjunct is linearized in its own Spell-Out domain, and

ordering statements established within each domain are preserved. SE_1 follows all other material in $Conj_A$; SE_2 follows all other material in $Conj_B$. After convergent deletion collapses the two into a single pronunciation event SE, that surviving event must satisfy the ordering constraints inherited from both domains. The only linear position satisfying both is the absolute right edge of the entire &P, deriving the Right-Edge Restriction without rightward movement. The converse prediction—left-edge convergence in head-final languages—is consistent with Left Node Raising in Korean and Japanese (Yatabe 2001; Kim et al. 2026).

Suspended Affixation provides independent support from within the same cell. In Meadow Mari, the conjunction *den* triggers obligatory deletion of identical inflectional material at the right edge of non-final conjuncts (Guseva & Weisser 2018):

(43) a. Me peres, pij den kajek-vlak-**em** už-am.
1SG cat dog and bird-PL-ACC see-1SG.PRES
'I see cats, dogs and birds.'

b. * Me peres-em den pij-vlak-em už-am.
1SG cat-ACC and dog-PL-ACC see-1SG.PRES

The CCT delivers the same verdict here as in clausal RNR: identical affixal material in parallel positions cannot be selectively retained or deleted; convergent deletion is the sole uniform outcome.

The CCT does not adjudicate between structural implementations of RNR. Its contribution is orthogonal: it provides the algebraic trigger for the uniformity requirement that all implementations—multidominance (Abels 2004; Wilder 2008), two-copy deletion (Belk et al. 2023), or PF Pruning—must satisfy but have previously stipulated.

RNR therefore fills the PF-identical cell in exactly the way the typology predicts. Uniformity is imposed by the CCT; convergence rather than distribution follows from identity; and right-edge placement follows from Cyclic Linearization.

To conclude, the grammar, unlike Buridan's ass, does not starve before symmetric

alternatives; it just computes exhaustively.

5 Interpreting Symmetry

PF and NS each possess mechanisms to restructure a symmetric input: post-syntactic trivialization and narrow-syntactic dissolution respectively (§4). LF has no such restructuring mechanism. Compositional rules operate over the transferred domain as such, and the underdetermination surfaces as interpretive instability: multiple compositional modes are in principle compatible with the transferred structure. The CCT constrains which of these survive, excluding asymmetric parses while leaving a unique σ-invariant rule available. Given independently motivated type-driven semantics, the surviving parse is conjunction.

5.1 LF Equivariance and the Conjunction Parse

As established in §3, the symmetric root complex $\{\sqrt{A}, \sqrt{B}\}$ reaches the interfaces lacking hierarchical asymmetry and formal labels. Bare roots serve merely as indices to encyclopedic concepts (Marantz 2001; Borer 2013).[35] As the syntactic representation fails to encode sufficient hierarchical information to specify a compositional mode (e.g., function-argument application), it admits multiple competing interpretations at LF (May 1985; Higginbotham 1985). The CCT acts as a negative algebraic filter over these candidates.

Insofar as semantic interpretation is homomorphic, mapping syntactic combination to semantic combination in a structure-preserving way (Montague 1970; Janssen 1997), it inherits equivariance. This yields an LF-internal version of the CCT grounded in compositionality alone:

(44) **LF CCT (Theorem of Compositionality):**

[35] See Dixon (2004); Francez & Koontz-Garboden (2017); Beavers & Koontz-Garboden (2020); Yu et al. (2023) for inherent-typing traditions. The robustness of the CCT result across typing implementations is noted at the end of this subsection.

Let h be the compositional interpretation function and $\sigma \in \mathrm{Aut}(\mathfrak{D})$ an automorphism of a syntactic domain $\mathfrak{D}$. If h is a homomorphism, then $h(\sigma(\mathfrak{D})) = \sigma'(h(\mathfrak{D}))$ for some automorphism σ' of the semantic domain. Therefore, if $\alpha \approx_{\mathrm{LF}} \beta$, any compositional rule must assign α and β automorphism-equivalent semantic values.[36]

Applying the transposition $\sigma = (\sqrt{A}\ \sqrt{B})$ to each candidate compositional operation and checking for σ-invariance, the theorem eliminates asymmetric compositional modes; it does not by itself choose among the surviving invariant candidates:

(45) σ-invariance test on candidate compositional operations for $\{\sqrt{A}, \sqrt{B}\}$:

a. *Modification* ($\sqrt{A}$ modifies $\sqrt{B}$, or vice versa): Applying σ reverses the dependency, creating a distinct semantic object. This violates σ-invariance and is ruled out by the CCT.[37]

b. *Disjunction* ($\mathrm{Concept}_A \vee \mathrm{Concept}_B$): σ-invariant. Excluded by independent architectural considerations (see below).

c. *Conjunction* ($\mathrm{Concept}_A \wedge \mathrm{Concept}_B$): σ-invariant. Licensed as the type-driven default.

With asymmetric parses eliminated, the selection of conjunction over disjunction among the remaining σ-invariant candidates follows from structural configuration. Predicate Modification (PM) is a default rule triggered by structure alone: two same-type

[36]The proof is straightforward. Let $\sigma = (\alpha\ \beta) \in \mathrm{Aut}_{\mathrm{LF}}(\mathfrak{D})$. By compositionality, $h(\mathfrak{D})$ depends only on the denotations and structural relations within $\mathfrak{D}$. Since σ preserves all structural relations, $h(\mathfrak{D}) = h(\sigma(\mathfrak{D}))$. Suppose h assigns distinct values: $h(\alpha) = x \neq y = h(\beta)$. Then $h(\sigma(\alpha)) = h(\beta) = y$ and $h(\sigma(\beta)) = h(\alpha) = x$, contradicting $h(\mathfrak{D}) = h(\sigma(\mathfrak{D}))$. Therefore h must treat α and β uniformly. Note that the proof holds specifically for sisters whose structural positions are interchangeable under σ, where neither is designated as functor or argument. In standard asymmetric configurations (e.g., $\{H, \mathrm{XP}\}$), the automorphism group is trivial and the equivariance condition is vacuously satisfied, imposing no constraint on composition.

[37]As Set-Merge is unordered, both directions of modification are theoretically possible despite the constant right-headed manifestation in Sinitic (Starosta et al. 1998). The CCT rules out both simultaneously.

sisters under a single node (Heim & Kratzer 1998). Pointwise disjunction, by contrast, requires a lexically introduced operator ($\vee$) and is not a default composition rule in standard type-driven frameworks.[38]

The categorizer a takes scope over $\{\sqrt{A}, \sqrt{B}\}$ and maps the acategorial roots to predicates of type $\langle e, t\rangle$. This uniform categorization is itself a CCT effect: as a deterministic function over a symmetric set, a cannot selectively categorize one root over the other. LF then faces two sibling $\langle e, t\rangle$ predicates within a single projection. As Functional Application requires sisters of type $\langle \alpha, \beta\rangle$ and α, it fails, leaving PM as the sole convergent composition rule: $[\![\text{PM}]\!] = \lambda x.\, [\![\alpha]\!](x) \wedge [\![\beta]\!](x)$.[39]

The result is therefore a compatibility theorem: the CCT constrains which compositional rules can apply; standard type-driven semantics selects among the survivors. Default PM is contingent on higher functional structure (InstP or AspP) establishing a single event/state variable. Without it, LF cannot determine whether the symmetric set constitutes a unified conjunctive predicate or two independent elements awaiting further composition. This is why bare *AB, lacking the requisite higher functional structure, crashes at C-I even though PM is in principle available as a composition rule.

(46) **Default PM:**

For a symmetric root set $\{\sqrt{A}, \sqrt{B}\}$ dominated by a categorizing head (a or v) establishing a single predicational domain (InstP or AspP):

a. The categorizer maps over the symmetric set, yielding a set of typed predicates:

[38] Under radical alternative semantics (Charlow 2014, 2019), the phase-theoretic architecture independently forces conjunction. PM at the categorizing phase boundary simultaneously fixes root meaning (contextual allosemy; Marantz 2013) and closes the alternative set, yielding conjunction. Disjunction would require alternatives to escape the phase unclosed. This formal necessity converges with cross-linguistic observations that conjunction is universally the unmarked operation relative to disjunction (Gil 1991; Haspelmath 2007).

[39] The same result is also directly derivable under Predicate Conjunctivism (Pietroski 2005, 2018), which posits conjunction as the default reflex of Merge.

$$[\![a\ \{\sqrt{A}, \sqrt{B}\}]\!] = \{\lambda x.\text{Concept}_A(x),\ \lambda x.\text{Concept}_B(x)\}$$

b. Within the aP projection, the two sister roots combine via PM:

$$[\![a\text{P}]\!] = \lambda x.\big[\text{Concept}_A(x) \wedge \text{Concept}_B(x)\big]$$

This result is robust across root-typing traditions. Under inherent-typing approaches where roots bear compositional types prior to categorization (Francez & Koontz-Garboden 2017; Yu et al. 2023), all Sinitic property-concept roots share type $\langle e, \langle s, t\rangle\rangle$. Function Application cannot selectively apply, and PM remains the unique σ-invariant rule at the symmetric node.

The same algebraic condition extends to phrasal coordination. The syntactic derivation of ATB uniformity in §4.4.1 proceeded from Evaluator Determinism; the present LF route proceeds from compositionality and homomorphic equivariance. That two derivations proceeding from independent architectural premises converge on the same empirical prediction—obligatory uniform treatment of symmetric conjuncts—supports the domain-generality of the restriction.

5.2 Two Structures, Two Interpretations

The distinct interpretations of AABB and ABAB follow compositionally from their respective functional spines, confirming that the two patterns are not phonological variants but encode distinct eventuality structures.[40] The same conjunctive mechanism survives in both structures, but distinct functional spines determine whether the conjoined properties are evaluated over a single state or distributed across sub-events.

In the AABB derivation, the stative head Inst introduces a single state variable s scoping over both predicates simultaneously:

(47) $[\![\text{InstP}_{\text{AABB}}]\!] = \lambda x.\exists s\big[\text{State}(s) \wedge \text{Theme}(s, x) \wedge \text{Sweet}(s) \wedge \text{Salty}(s)\big]$

[40] Syntactic symmetry in the present sense must be distinguished from *semantic symmetry* (Mayr & Schmitt 2017; Sportiche 2024), defined as truth-conditional invariance under conjunct permutation. The two notions are logically independent.

Since both roots predicate of the identical state s, simultaneous conjunction forces spatial or material distribution, deriving the zebra diagnostic of §2.

In the ABAB derivation, the verbalizer v^0 and Asp introduce an event variable e with an iterative structure, distributing predicates across ordered sub-events (Krifka 1990; Landman 2000; Champollion 2016):[41]

(48) $$\llbracket \text{AspP}_{\text{ABAB}} \rrbracket = \lambda x.\exists e\big[\text{Iter}(e) \wedge \text{Theme}(e,x) \wedge \text{Part}(e) = \{e_1, e_2, \ldots, e_n\}$$
$$\wedge\, \forall e_i \in \text{Part}(e).\, [\text{Sweet}(e_i) \vee \text{Salty}(e_i)]$$
$$\wedge\, \exists e_j \in \text{Part}(e)[\text{Sweet}(e_j)] \wedge \exists e_k \in \text{Part}(e)[\text{Salty}(e_k)]\big]$$

The contextual complementarity between AABB and ABAB independently confirms the distinction:

(49) a. Static Context (Constant Perception):
zhe-kuai binggan you-tou-dao-wei chi-qilai dou-shi [*tian-tian-xian-xian*
this-CL biscuit from-head-to-tail eat-INCH all-COP sweet-sweet-salty-salty
/ # *tian-xian-tian-xian*] *de*
/ sweet-salty-sweet-salty DE
'From beginning to end, this biscuit consistently tastes sweet and salty.'

b. Dynamic Context (Emergent Alternation):
ni jiao duo yi-hui, *manman jiu-hui chi-dao* [
you chew more one-while, slowly then-will eat-RES
tian-xian-tian-xian / # *tian-tian-xian-xian*] *de ceng-ci*
sweet-salty-sweet-salty / sweet-sweet-salty-salty DE layer
'If you chew longer, you gradually taste the alternating layers of sweetness and saltiness.'

AABB resolves indeterminacy by collapsing properties into a single state; ABAB resolves it by distributing them across event structure. The same Default PM, applied within different functional spines, deterministically yields different truth conditions.

[41] Part(e) denotes the aspectually determined partition of e into maximal homogeneous sub-events, following Krifka (1990, 1998).

5.3 Conceptual Integrability: A Limited Interface Filter

The DC is not a comprehensive theory of conjunction or conceptual compatibility. Its role in the present argument is restricted: it differentiates dynamically generated coordinatives from stored Listemes (§4.3.2) and constrains the range of roots eligible for dynamic coordination. Roots denoting abstract or cross-dimensional properties are systematically excluded from the coordinative pattern, though they are freely coordinated phrasally (§2).

(50) **Dimensionality Constraint (DC):**

$\textsc{Conj}(\sqrt{A}, \sqrt{B})$ converges at C-I iff the encyclopedic sortal categories $S(A)$ and $S(B)$ assigned by the Encyclopedia satisfy $S(A) \cap S(B) \neq \emptyset$.[42]

The intuition is straightforward. Default PM conjoins two predicates within a single categorizing phase without an overt coordinator to mediate presupposition accommodation. When the sortal domains of the two roots are disjoint (e.g., $\textsc{Color} \cap \textsc{Dimension} = \emptyset$), the conjoined predicate forces an empty extension, an instance of L-analytic deviance (Gajewski 2002; Chierchia 2013). In phrasal coordination, by contrast, an overt coordinator independently accommodates each conjunct's domain restriction, so cross-dimensional conjunction is freely available.

(51) Convergent: *tian-tian-suan-suan* ('sweet-sweet-sour-sour')

$D_{\textsc{taste}} \cap D_{\textsc{taste}} = D_{\textsc{taste}} \neq \emptyset \quad \checkmark$

(52) Crash: **hung-hung-da-da* ('red-red-big-big')

$D_{\textsc{color}} \cap D_{\textsc{dimension}} = \emptyset \quad \perp$

Monosyllabic AA reduplication confirms the C-I locus of this restriction. While **hung-hung-da-da* crashes, the independently reduplicated forms *hung-hung de, da-da de* are fully acceptable as separate predicates. Neither NS nor PF excludes the roots; the derivation crashes at C-I because the DC is not satisfied.

[42] $S(\cdot)$ maps a root to its encyclopedic property-type classification (e.g., Color, Taste), not to the set of entities it predicates over.

Cross-dialectal variation further supports an encyclopedic locus. Cantonese enforces a strict uni-modal DC, excluding cross-modal combinations such as **hung4-hung4-laat6-laat6* ('red-red-spicy-spicy'). Mandarin evaluates the DC more permissively, licensing the equivalent *hong-hong-la-la*. Both varieties nonetheless exclude abstract-dimensional combinations such as **hong-hong-da-da*. The parameterization is internal to the Encyclopedia, consistent with the Borer-Chomsky Conjecture (Borer 1984, 2013) about where variation resides.

The DC generates a directly testable prediction for scalar binding. Degree modification presupposes a single scalar dimension (von Stechow 1984; Kennedy 1999, 2007). A lexicalized compound maps onto a monodimensional Gestalt, providing a single scale for Deg to bind. A coordinative, by contrast, yields a multidimensional conjunctive predicate, and a single Deg head cannot simultaneously quantify over disjoint scales. Cantonese provides the minimal pair:

(53) *ne1-go3 dei6-fong1 zan1-hai3* [*hou2 gon1-gon1-zeng6-zeng6*] *wo3*
this-CL place really very clean-clean-purified-purified SFP
'This place is indeed very clean.'

(54) * *ni1-faai3 beng2 zan1-hai3* [*hou2 tim4-tim4-haam4-haam4*]
this-CL biscuit really very sweet-sweet-salty-salty
Intended: 'This biscuit is very sweet and very salty.'

The listed compound *gon1-zeng6* ('clean') supports degree modification; the coordinative *tim4-haam4* ('sweet-salty') does not. This independently validates the Three-Routes partition (§4.3.2): stored Listemes preserve a monodimensional Gestalt compatible with scalar binding, while dynamically generated coordinatives yield multidimensional predicates that resist it.

6 Concluding Remarks

6.1 An Automorphic Grammar

The deterministic evaluators of the grammar can be uniformly characterized as computations whose behavior is fully determined by the automorphism structure of their input domains:

(55) **Automorphism Sensitivity:**

For any deterministic evaluator $\mathcal{A}$ operating over a syntactic domain $\mathfrak{D}$:

a. If $\mathrm{Aut}_{\mathcal{A}}(\mathfrak{D})$ is trivial (only the identity), $\mathcal{A}$ computes a unique output over $\mathfrak{D}$.

b. If $\mathrm{Aut}_{\mathcal{A}}(\mathfrak{D})$ is non-trivial, $\mathcal{A}$ can only return properties invariant under every element of $\mathrm{Aut}_{\mathcal{A}}(\mathfrak{D})$. If no such invariant properties exist, $\mathcal{A}$ has no well-defined output.

This single property underlies every result in the paper: the Null Label as a fourth labeling case (§3.2.2); the *AB gap; the obligatory AABB and ABAB resolutions as trivialization and dissolution respectively (§4); the convergence of stored compounds via encyclopedic pre-emption (§4.3.2); and the 2×2 typology unifying these with ATB extraction and RNR (§4.4). At LF, the same restriction admits an independent derivation grounded in compositionality alone (§5.1): the syntactic route proceeds from Evaluator Determinism, the interpretive route from homomorphic equivariance, and the two converge on the same empirical prediction—obligatory uniform treatment of symmetric domains—from independent architectural premises.

This convergence carries an architectural consequence worth making explicit. Prominent accounts of ATB appeal to Sideward Movement or Multidominance; prominent accounts of RNR appeal to rightward movement or stipulated uniformity; prominent accounts of Sinitic reduplication invoke construction-specific templates. The present account derives all four cells of the typology from Set-Merge, Evaluator Determinism, and

independently motivated interface conditions. The architectural compression achieved by the proposal can be summarized as follows.

Introduced	**No longer required**
Evaluator Determinism (named, not new)	Stipulated ATB uniformity
CCT (theorem, not mechanism)	Rightward movement for RNR
DEC (corollary of CCT + Stray Affix)	Construction-specific reduplication templates
Algorithm-relative indistinguishability	Sideward Movement / Multidominance (for symmetric cases)

Table 9: Architectural compression: what is introduced vs. what is replaced.

The framework generates programmatic predictions beyond the present empirical scope. Under the Copy Theory of Movement (Chomsky 1995; Nunes 2004), copy deletion should require computable asymmetry between copies, correctly deriving double pronunciation when copies occupy genuinely symmetric positions (Potsdam & Runner 2001; cf. Malanoski & Kandybowicz 2026). Similar predictions extend to any grammatical operation requiring a choice between indistinguishable elements, including Closest Conjunct Agreement and binding under symmetric c-command. The concrete derivations are deferred to future work.

The automorphism-based framework converges with independent work by Frank et al. (to appear), who define the class NoAut, trees admitting no non-trivial automorphism, and prove that NoAut imposes restrictions similar to those derived from the LCA. Where their approach identifies *which* structures are well-formed, the present approach explains *why*: deterministic algorithms cannot operate over domains whose automorphism groups are non-trivial and whose invariant sets are empty. The grammar does not 'care about' symmetry *per se*; it just computes deterministically, and symmetry is the algebraic name for when deterministic computation is underdetermined.

6.2 Epiphenomenal Convergence

The Sinitic adjectival coordinatives provide strong empirical support for the SMT and a strictly modular architecture. The AABB and ABAB patterns do not originate from a dedicated grammatical rule or a phonological template, but arise from the intersection of independently motivated interface requirements.

The phenomenon confirms that NS does not filter out ill-formed structures. The Sinitic pattern demonstrates something that goes beyond Ott's (2015) claim that NS freely generates structures that may fail at the interfaces: NS can generate structures that are *locally* ill-formed—failing labeling, resisting linearization, admitting no unique compositional interpretation—yet *globally* convergent once the interfaces independently impose their respective conditions.

The obligatory doubling of *AB highlights the active yet independent role of the interfaces. PF and LF each resolve symmetry in their own domain, and neither has access to the other's operations. The surface regularity reflects the alignment of distinct interface conditions, an empirical instantiation of the Configurational Convergence corollary (22): convergence is a global property, jointly determined by every evaluator at every interface, yet the alignment that delivers it is purely epiphenomenal.

Several boundaries should be acknowledged. The CCT derives the symmetric component of the Coordinate Structure Constraint; the asymmetric component falls outside its algebraic scope and requires independent locality-theoretic explanation (cf. Bošković 2020a; Sportiche 2024). The paper does not adjudicate among competing structural implementations of RNR; it constrains what any such implementation must satisfy. The Dimensionality Constraint is a restricted encyclopedic filter, not a general theory of conjunction. And the broader predictions noted above—copy deletion, Closest Conjunct Agreement, binding—remain programmatic. The present account covers the symmetric case; non-

symmetric coordination phenomena require independent treatment.

The CCT is bidirectional as a diagnostic. Forward: non-trivial Aut predicts exhaustive effects. Backward: obligatory exhaustive effects diagnose non-trivial Aut. The grammar's behavioral surface thus provides an indirect window onto the automorphism structure of its underlying domains.

6.3 From Breaking to Translation

Since Moro's formulation of Dynamic Antisymmetry, it has been widely assumed that symmetric configurations are inherently unstable and must be resolved prior to Transfer. The Sinitic data suggest a different perspective. Pure root symmetry $\{\sqrt{A}, \sqrt{B}\}$ can, under specific morphosyntactic conditions, persist throughout the syntactic derivation and receive well-formed interpretations at the interfaces. Symmetry need not always be eliminated within NS, but may instead be tolerated and subsequently rendered legible through independent interface conditions.

Well-formedness is not enforced at a single locus but emerges as a distributed property, jointly determined by the independent constraints of NS, PF, LF, and C-I. A single underlying symmetric structure is systematically mapped onto distinct interface-specific representations, with the underlying symmetry preserved rather than eliminated—a process of *Interface Translation*. The reduplicated pattern is a morphological reflex of syntactic symmetry: a faithful externalization of a structural tie that the grammar lacks the internal resources to break.

To conclude, when a language lacks the internal syntax to break a tie, it keeps it, relying entirely on the active sensorimotor and conceptual systems to faithfully translate this symmetry into human speech and thought.